MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONTROLLER, INTERRUPT, DIGITAL, KEY CODE ZJX

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interrupt controller module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the *Department of Defense Index of Specifications and Standards* specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.297 failures per million operating hours for class I or 0.964 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389 When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ZJX.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/305-1 for class I and M28787/305-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.

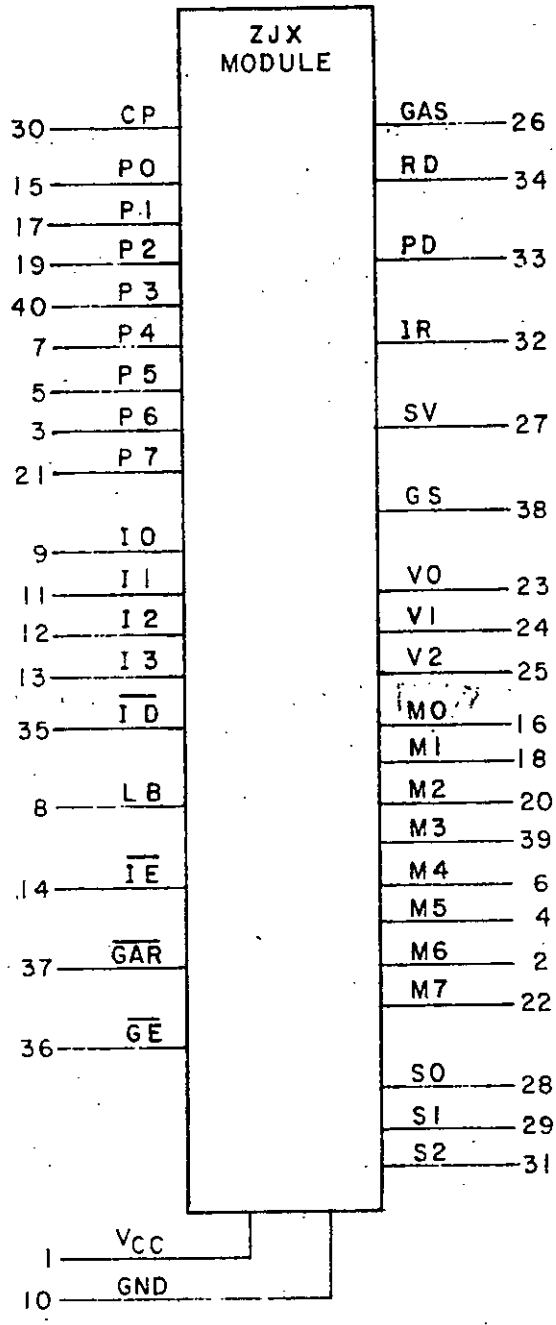


FIGURE 1. ZJX module function diagram, interrupt controller.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	Voltage (V_{CC}) stress test	III	S	← Must pass remaining tests →										
C1-C29	Input breakdown current	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D18	Input clamp diode voltage	V	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E19	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F14	Input current, high level	VII	X	0	20	0	20	0	20	0	20	0	20	µA
F15-F20	Input current, high level	VII	X	0	40	0	40	0	40	0	40	0	40	µA
F21-F29	Input current, high level	VII	X	0	60	0	60	0	60	0	60	0	60	µA
G1-G16	Input current, low level	VIII	X	0.0	-0.87	0.0	-0.87	0.0	-0.87	0.0	-0.87	0.0	-0.87	mA
G17	Input current, low level	VIII	X	0.0	-0.433	0.0	-0.433	0.0	-0.433	0.0	-0.433	0.0	-0.433	mA
G18	Input current, low level	VIII	X	0.0	-2.17	0.0	-2.17	0.0	-2.17	0.0	-2.17	0.0	-2.17	mA
G19-G21	Input current, low level	VIII	X	0.0	-0.108	0.0	-0.108	0.0	-0.108	0.0	-0.108	0.0	-0.108	mA
G22-G29	Input current, low level	VIII	X	0.0	-0.163	0.0	-0.163	0.0	-0.163	0.0	-0.163	0.0	-0.163	mA
H1,H2, H4-H20	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H3	Output voltage, high level	IX	X	2.4	4.55	2.4	4.55	2.4	4.55	2.4	4.55	2.4	4.55	V
J1-J20	Output voltage, low level	X	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K3	High-impedance leakage current, high level	XI	X	-1	50	-1	50	-1	50	-1	50	-1	50	µA
K4-K11	High-impedance leakage current, high level	XI	X	-1	150	-1	150	-1	150	-1	150	-1	150	µA
K12-K14	High-impedance leakage current, high level	XI	X	-1	100	-1	100	-1	100	-1	100	1	100	µA

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

★ TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	> 25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L1-L3	High-impedance leakage current, low level	XII	X	1	-50	1	-50	1	-50	1	-50	1	-50	µA
L4-L11	High-impedance leakage current, low level	XII	X	1	-150	1	-150	1	-150	1	-150	1	-150	µA
L12-L14	High-impedance leakage current, low level	XII	X	1	-100	1	-100	1	-100	1	-100	1	-100	µA
M	Output leakage current	XIII	X	0	250	0	250	0	250	0	250	0	250	µA
N	Power supply current	XIV	X	26	260	27	275	27	275	30	310	30	310	mA
P1-P6	Dynamic functional test	XV	X	← See table XV →										
Q1-Q5	Transition time (t_{TLH}), C_L = 50 pF	XVI	S	1	17	1	17	1	22	1	17	1	27	ns
R1-R6	Transition time (t_{THL}), C_L = 50 pF	XVII	S	1	6	1	6	2	11	1	6	2	16	ns
S1,S2	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	4	47	4	47	5	54	4	47	6	62	ns
S3	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	4	42	4	42	5	51	4	42	6	60	ns
S4	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	2	13	2	13	2	14	2	13	2	15	ns
S5	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	2	23	2	23	2	28	2	23	3	33	ns
S6	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	8	82	8	82	9	93	8	82	10	105	ns
S7, S8	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	5	57	5	57	6	66	5	57	7	75	ns
S9	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	6	66	6	66	7	75	6	66	8	85	ns
S10	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	4	42	4	42	4	48	4	42	5	54	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S11	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	4	45	4	45	5	51	4	45	5	58	ns
S12	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	3	32	3	32	3	38	3	32	4	45	ns
S13	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	5	56	5	56	6	61	5	56	6	66	ns
S14	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	2	25	2	25	3	30	2	25	3	35	ns
S15	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	8	82	8	82	9	93	8	82	10	105	ns
S16, S17	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	5	57	5	57	6	66	5	57	7	75	ns
S18	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	7	74	7	74	8	85	7	74	9	96	ns
T1, T2	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	4	47	4	47	5	54	4	47	6	62	ns
T3	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	4	42	4	42	5	51	4	42	6	60	ns
T4	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	2	13	2	13	2	14	2	13	2	15	ns
T5	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	2	23	2	23	2	28	2	23	3	33	ns
T6	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	8	82	8	82	9	93	8	82	10	105	ns
T7, T8	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	5	57	5	57	6	66	5	57	7	75	ns
T9	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	6	66	6	66	7	75	6	66	8	85	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° +5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T10	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	4	42	4	42	4	48	4	42	5	54	ns
T11	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	4	45	4	45	5	51	4	45	5	58	ns
T12	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	3	32	3	32	3	38	3	32	4	45	ns
T13	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	5	56	5	56	6	61	5	56	6	66	ns
T14	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	2	30	2	30	3	30	2	30	3	35	ns
T15	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	8	82	8	82	9	93	8	82	10	105	ns
T16, T17	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	5	57	5	57	6	66	5	57	7	75	ns
T18	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	7	74	7	74	8	85	7	74	9	96	ns
U1-U5	Output crosstalk, low level	XX	Q	0	1,000	0	1,000	0	1,000	0	1,000	0	1,000	mV
V1-V4,V6-V9	Output crosstalk, high level	XXI	Q	0	700	0	700	0	700	0	700	0	700	mV
V5	Output crosstalk, high level	XXI	Q	0	900	0	900	0	900	0	900	0	900	mV
W	Power supply transient amplitude	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply filter capacitance	XXIII	S	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	µF
Y1-Y13,Y15, Y16	Input capacitance	XXIV	Q	2	25	2	25	2	25	2	25	2	25	pF
Y14	Input capacitance	XXIV	Q	2	45	2	45	2	45	2	45	2	45	pF
Y17, Y18	Input capacitance	XXIV	Q	2	35	2	35	2	35	2	35	2	35	pF
Z1-Z8	Input/output capacitance	XXV	Q	2	50	2	50	2	50	2	50	2	50	pF
Z9-Z11	Input/output capacitance	XXV	Q	2	40	2	40	2	40	2	40	2	40	pF

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA1	Output capacitance	XXVI	Q	2	35	2	35	2	35	2	35	2	35	pF
AA2-AA4	Output capacitance	XXVI	Q	2	25	2	25	2	25	2	25	2	25	pF
AA5-AA9	Output capacitance	XXVI	Q	2	40	2	40	2	40	2	40	2	40	pF
BB1-BB16	Disable/enable recovery time (t_{PLZ}, t_{PZL}), C_L = 50 pF	XXVII	S	5	52	5	52	5	58	5	52	6	64	ns
BB18,BB20, BB22,BB24, BB26,BB28, BB30,BB32	Disable/enable recovery time (t_{PZH}), C_L = 50 pF	XXVII	S	5	57	5	57	5	63	5	57	6	69	ns
BB17,BB19, BB21,BB23, BB25,BB27, BB29,BB31	Disable/enable recovery time (t_{PHZ}), C_L = 50 pF	XXVII	S	6	70	6	70	7	76	6	70	7	82	ns
BB33-BB38, BB40,BB42, BB44-BB50, BB52,BB54, BB56	Disable/enable recovery time (t_{PLZ}, t_{PZL}, t_{PZH}), C_L = 50 pF	XXVII	S	5	59	5	59	6	65	5	59	7	72	ns
BB39,BB41, BB43,BB51, BB53,BB55	Disable/enable recovery time (t_{PHZ}), C_L = 50 pF	XXVII	S	6	65	6	65	7	71	6	65	7	78	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A, C through X, and BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A, C through X, and BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A and C through BB shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:35Z
Check the source to verify that this is the current version before use.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Power dissipation (no load)	1.71 max	1.71 max	W
Input voltage			
Low voltage	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High voltage	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V
Input current			
Low level (V_{IL} = 0.0 V)			
S0-S2	0.0 min, -0.108 max	0.0 min, -0.108 max	mA
M0-M7	0.0 min, -0.163 max	0.0 min, -0.163 max	mA
Latch bypass	0.0 min, -0.433 max	0.0 min, -0.433 max	mA
Interrupt disable	0.0 min, -2.17 max	0.0 min, -2.17 max	mA
All others	0.0 min, -0.87 max	0.0 min, -0.87 max	mA
High level (V_{IH} = 2.7 V)			
$\overline{IE}$, $\overline{GE}$, $\overline{GAR}$	0 min, 40 max	0 min, 40 max	µA
S0-S2	0 min, 40 max	0 min, 40 max	µA
M0-M7, $\overline{ID}$	0 min, 60 max	0 min, 60 max	µA
All others	0 min, 20 max	0 min, 20 max	µA
Output voltage			
Low level (I_{OL} = 12 mA)	0.0 min, 0.5 max	0.0 min, 0.5 max	V
High level (I_{OH} = -1.0 mA)	2.4 min, 4.5 max	2.4 min, 4.5 max	V
Output leakage current			
Open collector			
(V_{OH} = 5.5 V) IR	0 min, 250 max	0 min, 250 max	µA
Setup time (relative to clock low-to-high transition)			
S0-S2	15 min	15 min	ns
M0-M7	15 min	15 min	ns
P0-P7	15 min	15 min	ns
LB	20 min	20 min	ns
$\overline{IE}$	55 min	55 min	ns
I0-I3	t_{pWL} +40 min	t_{pWL} +40 min	ns
$\overline{GE}$	15 min	15 min	ns
$\overline{GAR}$	15 min	15 min	ns
$\overline{ID}$	42 min	42 min	ns
Holdtime (relative to clock low-to-high transition)			
S0-S2	10 min	10 min	ns
M0-M7	10 min	10 min	ns
P0-P7	8 min	8 min	ns
LB	0 min	0 min	ns
$\overline{IE}$	0 min	0 min	ns
I0-I3	0 min	0 min	ns
$\overline{GE}$	13 min	13 min	ns
$\overline{GAR}$	13 min	13 min	ns
$\overline{ID}$	0 min	0 min	ns
P0-P7 Hold time relative to LB	20 min	20 min	ns

6.5 Module function. The ZJX module is a high-speed, eight-bit priority interrupt unit that is cascadable to handle any number of priority interrupt request levels. The high-speed of the ZJX module makes it ideal for use in 2900 family microcomputer designs, but it can also be used with other MOS microprocessors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

6.5.1 Interrupt requests. The ZJX module receives interrupt requests on eight-interrupt input lines (P0-P7). A low level is a request. An internal latch may be used to catch pulses on these lines, or the latch may be bypassed so the request lines drive the edge-triggered interrupt register directly. An eight-bit mask register is used to mask individual interrupts. Considerable flexibility is provided for controlling the mask register. Requests in the interrupt register are ANDed with the corresponding bits in the mask register and the results are sent to an eight-input priority encoder, which produces a three-bit encoded vector representing the highest numbered input which is not masked.

6.5.2 Status register. An internal status register is used to point to the lowest priority at which an interrupt will be accepted. The contents of the status register are compared with the output of the priority encoder, and an interrupt request output will occur if the vector is greater than or equal to status. Whenever a vector is read from the ZJX module, the status register is automatically updated to point to one level higher than the vector read. (The status register can be loaded externally or read out at any time using the S pins.) Signals are provided for moving the status upward across devices (group advance send and group advance receive) and for inhibiting lower priorities from higher order devices (ripple disable, parallel disable, and interrupt disable). A status overflow output indicates that an interrupt has been read at the highest priority.

* 6.5.3 Instruction field. The ZJX module is controlled by a four-bit instruction field I0-I3. The command on the instruction lines is executed if $\overline{IE}$ is low and is ignored if $\overline{IE}$ is high, allowing the four-I bits to be shared with other devices.

6.5.4 Module use. The MIL-M-28787/301(EC), /304(EC), /305(EC), and /308(EC) (key codes ERD, ZJU, ZJX, and MJC) modules are a series of designs, implementing the 2900 family as building blocks to emulate microprogrammed computers or controllers.

* 6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and even when presented as hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and may be obtained upon request.

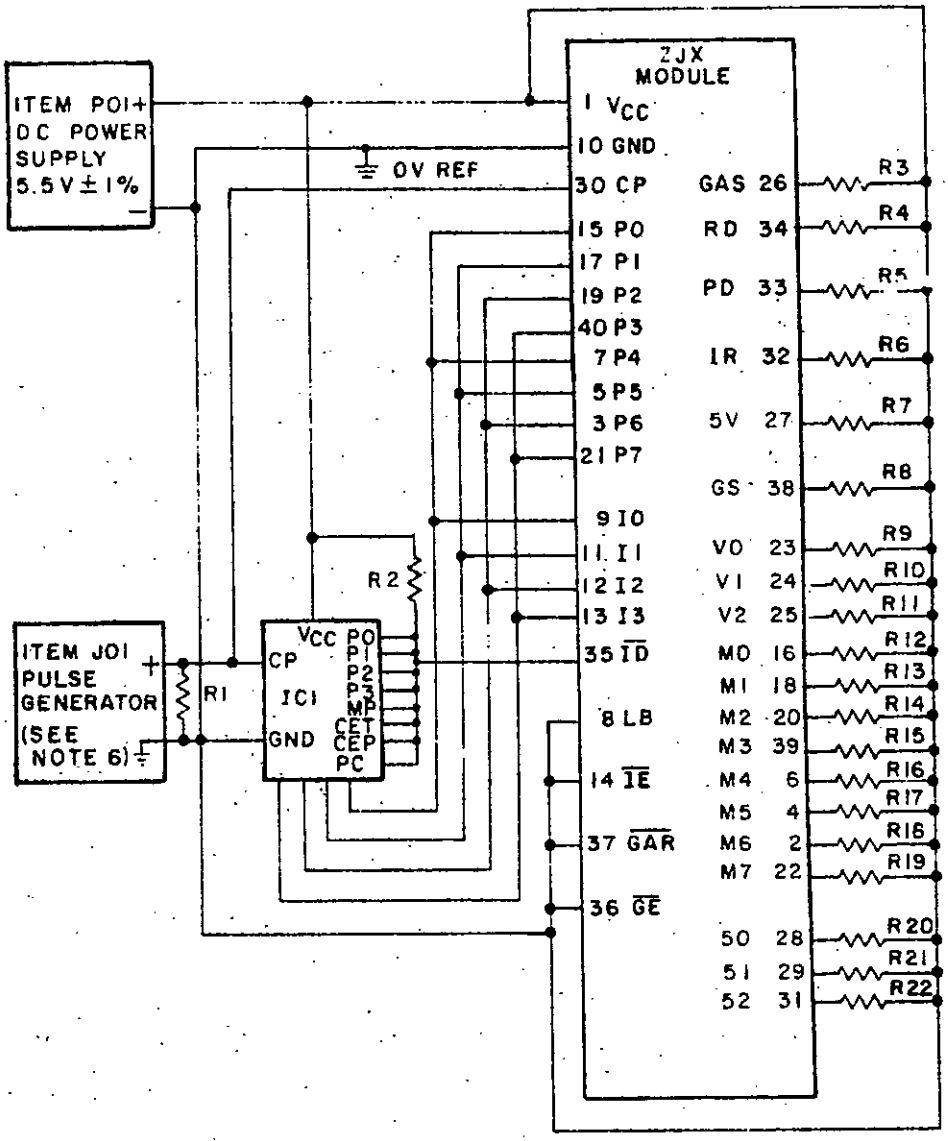
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N233)

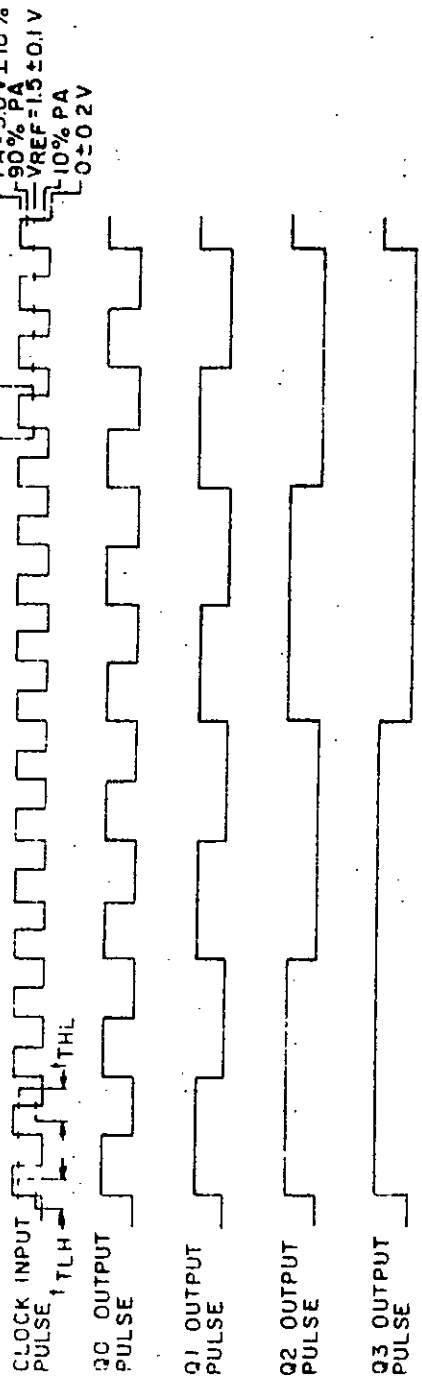
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. IC1 = 9316 or equivalent.
2. R1 = 51Ω ±5%, 1/2 W.
3. R2, R12-R22 = 1.2 kΩ ±5%, 1/4 W.
4. R3-R11 = 420Ω ±5%, 1/4 W.

* FIGURE 2. Operating temperature, component temperature, and life test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES - Continued.
5. CLK pulse = 100 kHz ±20% squarewave, duty cycle = 50% ±20%.
6. Item 001 setup:
   PW = 5.0 µs ±10%.
   $t_{TLH} = t_{THL}$ = 25 ±10 ns.
   50% duty cycle.

FIGURE 2. Operating temperature, component temperature, and life test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

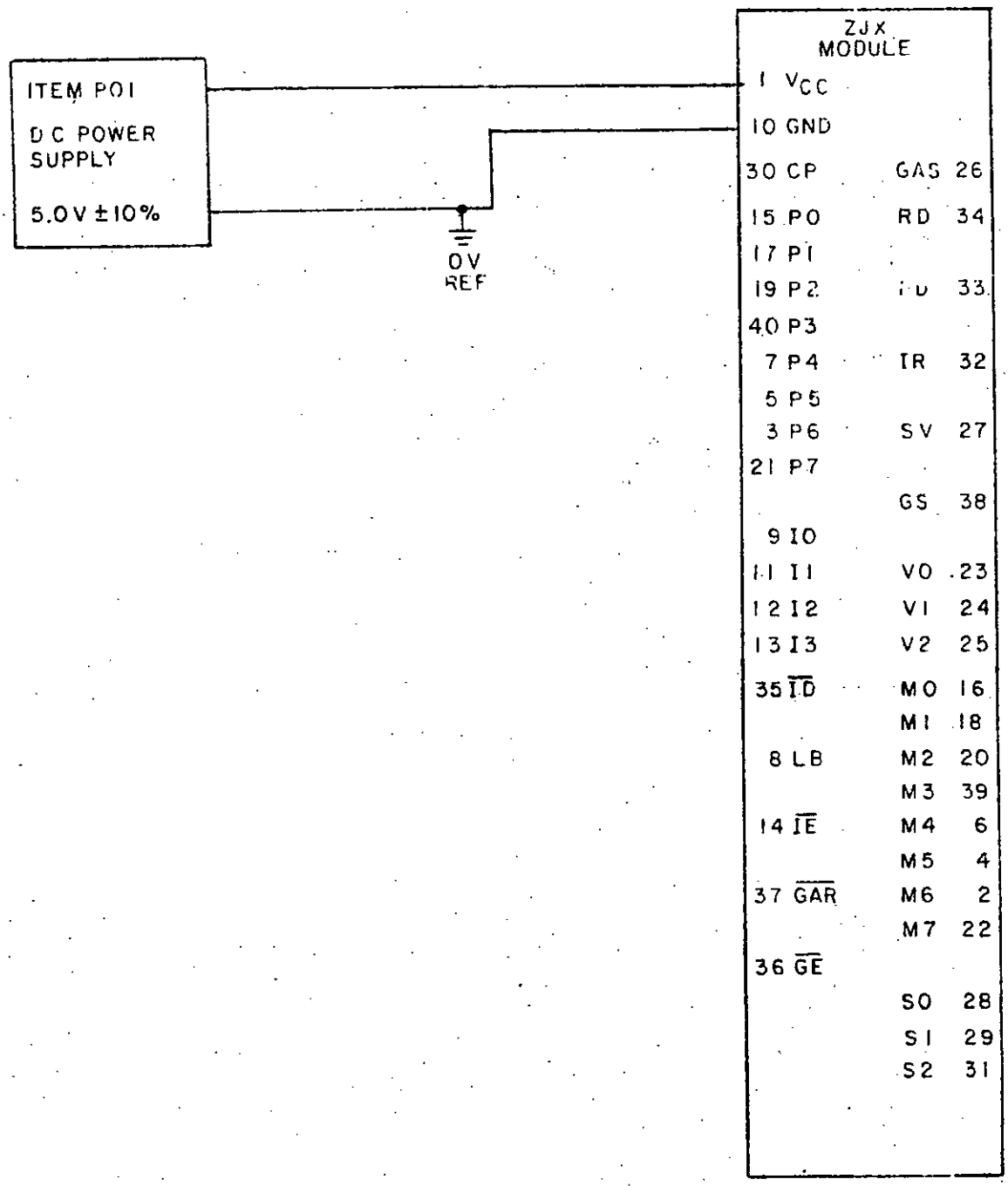


FIGURE 3. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

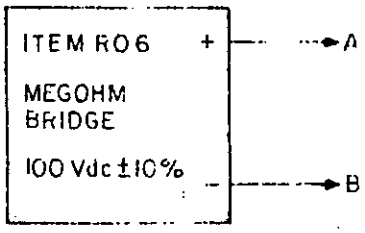
TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	4

TABLE III. Voltage (V_{CC}) stress test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
B	Voltage (V_{CC}) stress	← Must pass remainder of tests →				5

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.



WARNING:

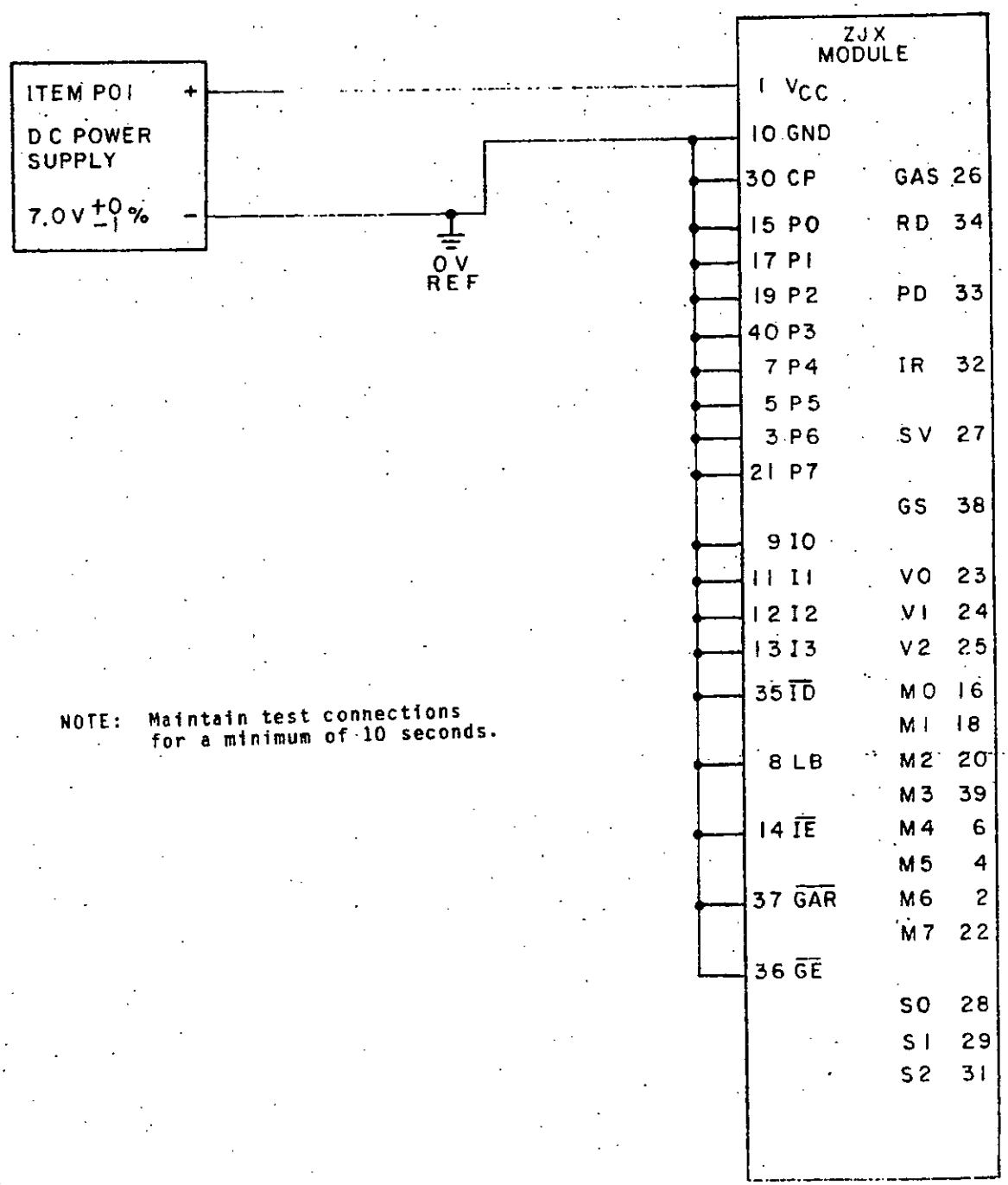
When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead.

ZJX MODULE			
1	V_{CC}		
10	GND		
30	CP	GAS	26
15	P0	RD	34
17	P1		
19	P2	PD	33
40	P3		
7	P4	IR	32
5	P5		
3	P6	SV	27
21	P7		
		GS	38
9	I0		
11	I1	V0	23
12	I2	V1	24
13	I3	V2	25
35	$\overline{ID}$	M0	16
		M1	18
8	LB	M2	20
		M3	39
14	$\overline{IE}$	M4	6
		M5	4
37	$\overline{GAR}$	M6	2
		M7	22
36	$\overline{GE}$		
		S0	28
		S1	29
		S2	31

FIGURE 4. Isolation test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.



NOTE: Maintain test connections for a minimum of 10 seconds.

FIGURE 5. Voltage (V_{CC}) stress test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE IV. Input breakdown current test.

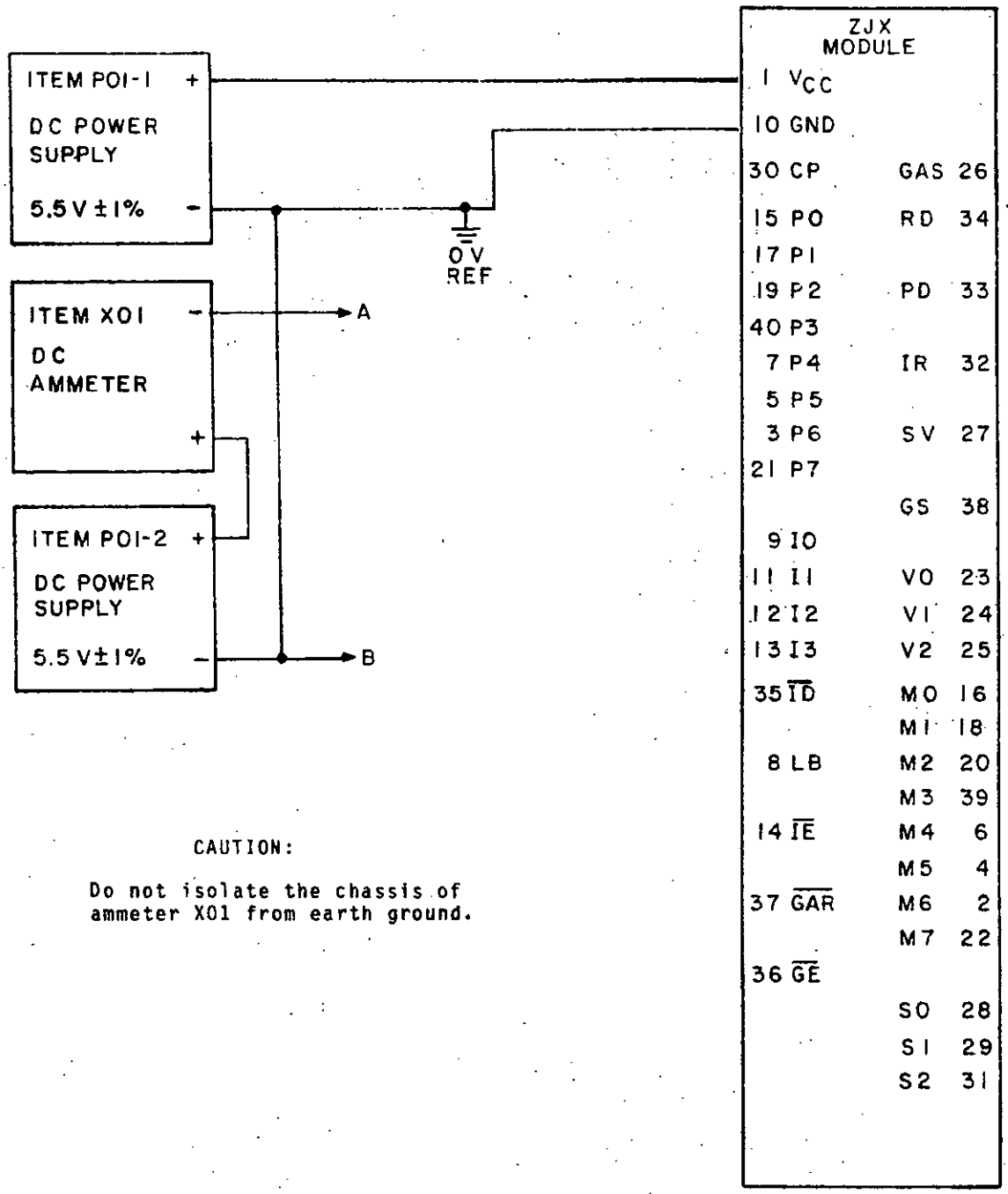
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	CP Input breakdown current	30	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,22,28,29,31,35,36,37,39,40	0.0	1.0	mA	6
C2	P0	15	2,3,4,5,6,7,8,9,11,12,13,14,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C3	P1	17	2,3,4,5,6,7,8,9,11,12,13,14,15,16,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C4	P2	19	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,20,21,22,28,29,30,31,35,36,37,39,40				
C5	P3	40	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39				
C6	P4	7	2,3,4,5,6,8,9,11,12,13,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C7	P5	5	2,3,4,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C8	P6	3	2,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C9	P7	21	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20,22,28,29,30,31,35,36,37,39,40				
C10	I0	9	2,3,4,5,6,7,8,11,12,13,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C11	I1	11	2,3,4,5,6,7,8,9,12,13,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C12	I2	12	2,3,4,5,6,7,8,9,11,13,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C13	I3	13	2,3,4,5,6,7,8,9,11,12,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				
C14	$\overline{ID}$	35	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,22,28,29,30,31,36,37,39,40				
15	LB	8	2,3,4,5,6,7,9,11,12,13,14,15,16,17,18,19,20,21,22,28,29,30,31,35,36,37,39,40				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C16	$\overline{IE}$ Input breakdown current	14	2,3,4,5,6,7,8,9,11,12,13,15,16,17,18, 19,20,21,22,28,29,30,31,35,36,37,39,40	0.0	1.0	mA	6
C17	$\overline{GAR}$	37	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,20,21,22,28,29,30,31,35,36,39,40				
C18	$\overline{GE}$	36	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,20,21,22,28,29,30,31,35,37,39,40				
C19	M0	16	2,3,4,5,6,7,8,9,11,12,13,14,15,17,18, 19,20,21,22,28,29,30,31,35,36,37,39,40				
C20	M1	18	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 19,20,21,22,28,29,30,31,35,36,37,39,40				
C21	M2	20	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,21,22,28,29,30,31,35,36,37,39,40				
C22	M3	39	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,20,21,22,28,29,30,31,35,36,37,40				
C23	M4	6	2,3,4,5,7,8,9,11,12,13,14,15,16,17,18, 19,20,21,22,28,29,30,31,35,36,37,39,40				
C24	M5	4	2,3,5,6,7,8,9,11,12,13,14,15,16,17,18, 19,20,21,22,28,29,30,31,35,36,37,39,40				
C25	M6	2	3,4,5,6,7,8,9,11,12,13,14,15,16,17,18, 19,20,21,22,28,29,30,31,35,36,37,39,40				
C26	M7	22	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,20,21,28,29,30,31,35,36,37,39,40				
C27	S0	28	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,20,21,22,29,30,31,35,36,37,39,40				
C28	S1	29	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,20,21,22,28,30,31,35,36,37,39,40				
C29	S2	31	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,20,21,22,28,29,30,35,36,37,39,40				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

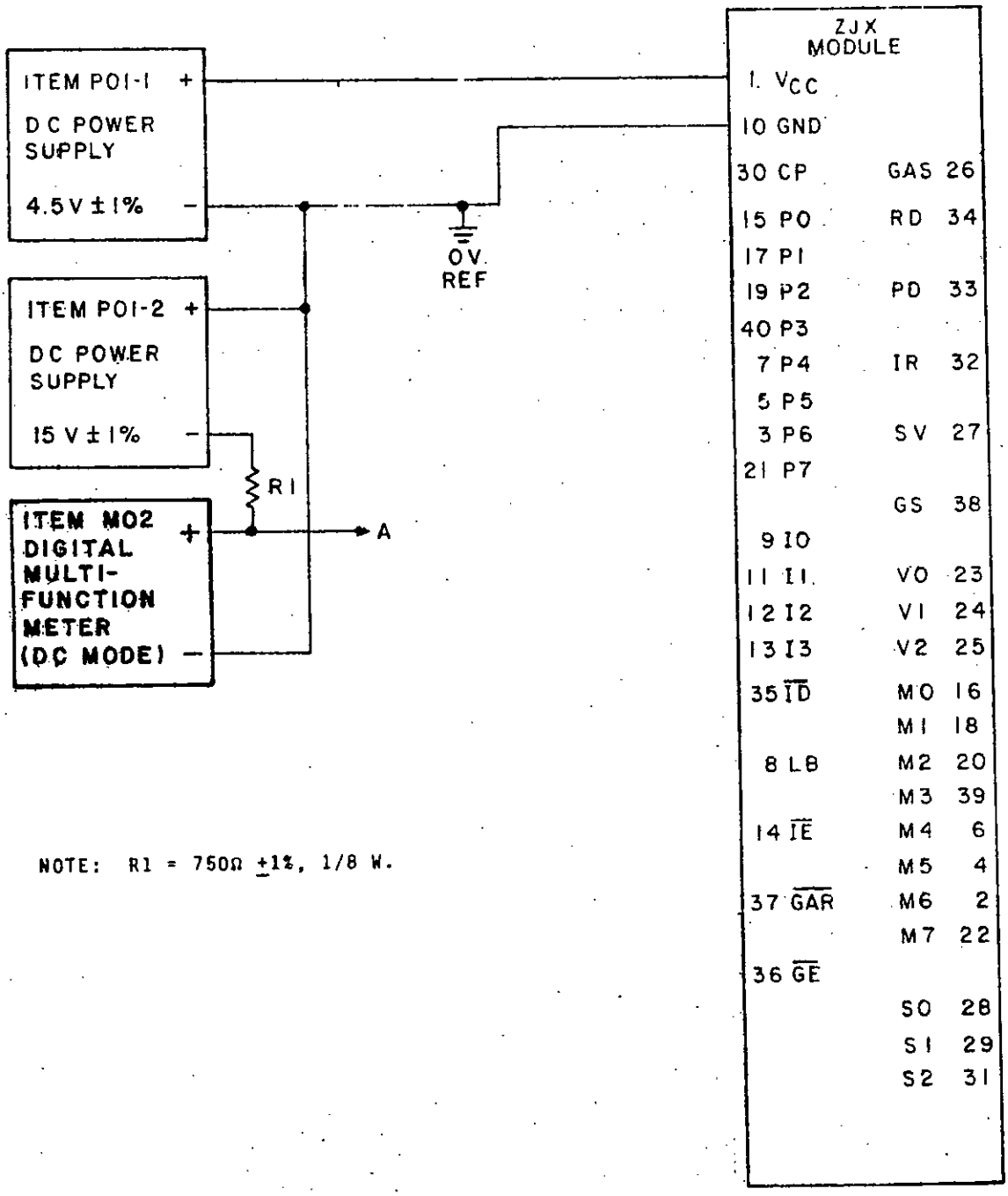
FIGURE 6. Input breakdown current test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	CP Input clamp diode voltage	30	-0.4	-1.5	V	7
D2	P0	15				
D3	P1	17				
D4	P2	19				
D5	P3	40				
D6	P4	7				
D7	P5	5				
D8	P6	3				
D9	P7	21				
D10	I0	9				
D11	I1	11				
D12	I2	12				
D13	I3	13				
D14	$\overline{ID}$	35				
D15	LB	8				
D16	$\overline{IE}$	14				
D17	$\overline{GAR}$	37				
D18	$\overline{GE}$	36				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 750Ω ±1%, 1/8 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test.

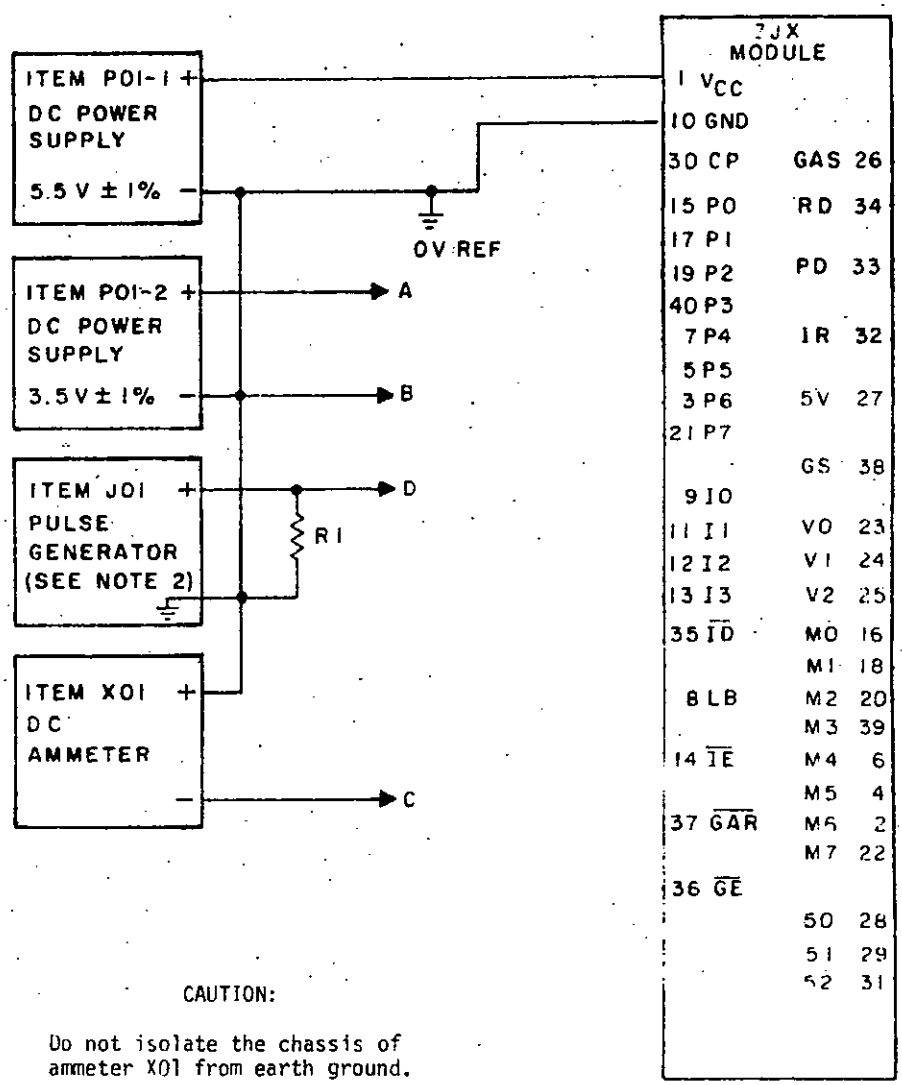
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	9,11,12,13,14	Open	30	---	---	---	8,9
E1	GAS Output short circuit current	Open	9,11,12,13,14	26	Open	-40	-100	mA	
---	Setup	9,12,21	11,13,14	Open	30	---	---	---	
E2	SV Output short circuit current	9,12,21	11,13,14	27	Open	-40	-100	mA	
---	Setup	Open	9,11,12,13,14	Open	30	---	---	---	
E3	GS Output short circuit current	↓	9,11,12,13,14	38	Open	-40	-100	mA	
E4	RD Output short circuit current		9,11,12,13,14	34	Open	-40	-100	mA	
---	Setup		9,11,12,13,14,36,37	Open	30	---	---	---	
E5	PD Output short circuit current		9,11,12,13,14,36,37	33	Open	-40	-100	mA	
---	Setup	3,5,7,9,12,15,17,19,40	11,13,14,21,37	Open	30	---	---	---	
E6	VO Output short circuit current	↓	↓	23	Open	-40	-100	mA	
E7	V1 Output short circuit current			24	Open	-40	-100	mA	
E8	V2 Output short circuit current			25	Open	-40	-100	mA	
---	Setup	13	9,11,12,14	Open	30	---	---	---	
E9	MO Output short circuit current	9,11,12	13,14	16	Open	-40	-100	mA	
E10	M1	↓	↓	18	↓	↓	↓	↓	
E11	M2			20					
E12	M3			39					
E13	M4			6					
E14	M5			4					
E15	M6			2					
E16	M7			22					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20 11:17:35Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	9,13,28,29, 31	11,12,14,36	Open	30	---	---	---	8,9
E17	S0 Output short circuit current	11,12	9,13,14,36	28	Open	-40	-100	mA	
E18	S1 Output short circuit current	11,12	9,13,14,36	29	Open	-40	-100	mA	
E19	S2 Output short circuit current	11,12	9,13,14,36	31	Open	-40	-100	mA	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



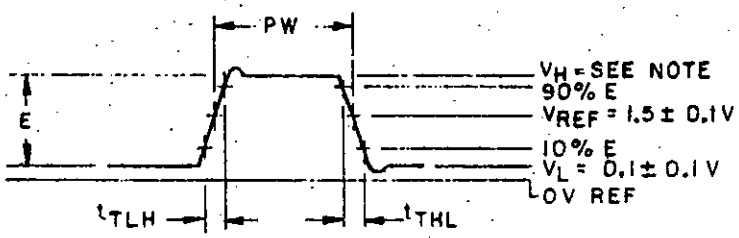
CAUTION:

Do not isolate the chassis of ammeter XOl from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. When point D is referenced, apply one clock pulse per figure 9 except for setup prior to test condition E2, then apply two clock pulses per figure 9.

FIGURE 8. Output short circuit current test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T17:35Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
V_H = 3.5 ±0.1 V for test conditions E1-E19, F1-F29, G1-G29 and N.
= 3.0 ±0.1 V for test conditions H1-H20, J1-J20, K1-K14, L1-L14 and M.
$t_{TLH} = t_{THL}$ = 10 ±5 ns.
PW = 500 ±50 ns.

FIGURE 9. Input pulse waveform for tables VI through XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current (high-level) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
---	Setup	9,11,12, 13,14	Open	Open	Open	30	---	---	---	10,9
F1	CP Input current, high level	Open		30		Open	0	20	µA	
F2	P0			15						
F3	P1			17						
F4	P2			19						
F5	P3			40						
F6	P4			7						
F7	P5			5						
F8	P6			3						
F9	P7			21						
F10	I0			9						
F11	I1			11						
F12	I2			12						
F13	I3			13						
F14	LB			8						
F15	$\overline{IE}$			14				40		
F16	$\overline{GAR}$			37				40		
F17	$\overline{GE}$			36				40		
---	Setup	11,12,14	9,13	Open		30	---	---	---	
F18	S0 Input current, high level	Open	Open	28		Open	0	40	µA	
F19	S1 Input current, high level	Open	Open	29		Open	0	40	µA	

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F20	S2 Input currrent, high level	Open	Open	31	Open	Open	0	40	µA	10,9
---	Setup	9,11,12, 13,14	↓	Open	↓	30	---	---	---	↓
F21	$\overline{ID}$ Input current, high level	Open	↓	35	↓	Open	0	60	µA	↓
---	Setup	9,14	11,12,13	Open	↓	30	---	---	---	↓
F22	M0 Input current, high level	Open	Open	16	↓	Open	0	60	µA	↓
F23	M1	↓	↓	18	↓	↓	↓	↓	↓	↓
F24	M2	↓	↓	20	↓	↓	↓	↓	↓	↓
F25	M3	↓	↓	39	↓	↓	↓	↓	↓	↓
F26	M4	↓	↓	6	↓	↓	↓	↓	↓	↓
F27	M5	↓	↓	4	↓	↓	↓	↓	↓	↓
F28	M6	↓	↓	2	↓	↓	↓	↓	↓	↓
F29	M7	↓	↓	22	↓	↓	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (low-level) test.

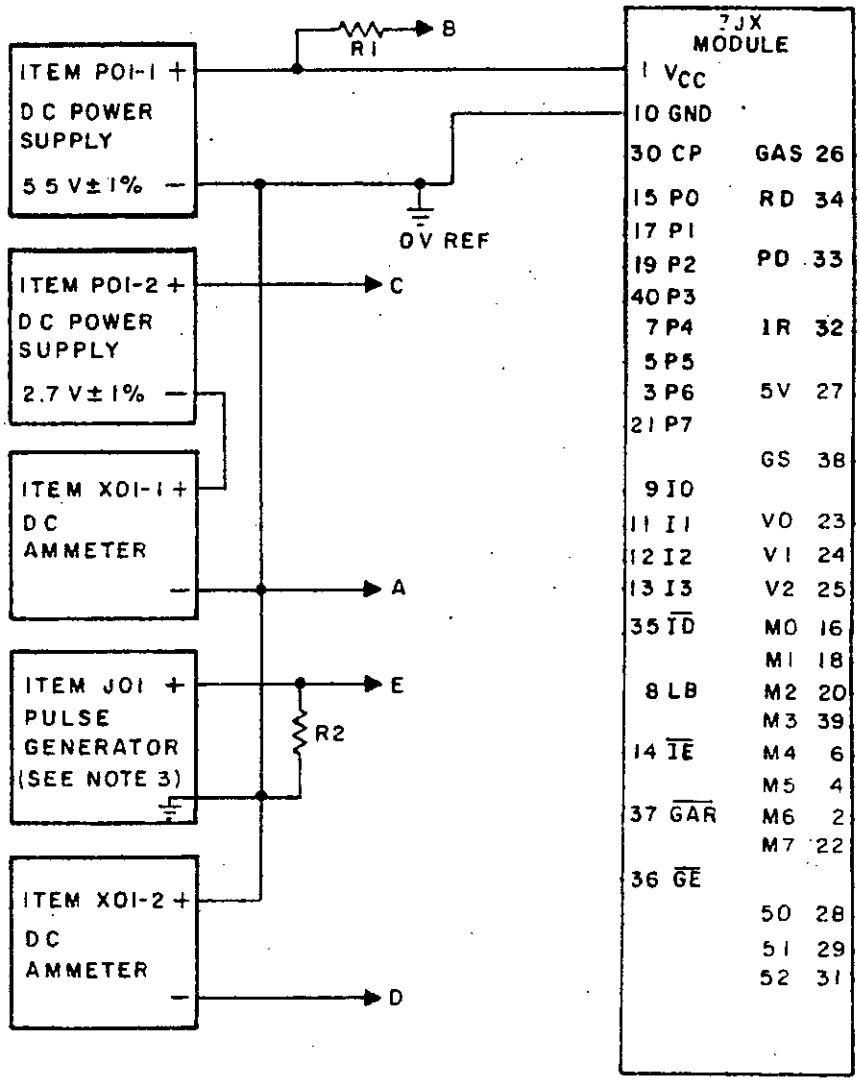
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
---	Setup	9,11,12, 13,14	Open	Open	Open	30	---	---	---	10,9
G1	CP Input current, low level	Open			30	Open	0.0	-0.87	mA	
G2	P0				15					
G3	P1				17					
G4	P2				19					
G5	P3				40					
G6	P4				7					
G7	P5				5					
G8	P6				3					
G9	P7				21					
G10	I0				9					
G11	I1				11					
G12	I2				12					
G13	I3				13					
G14	$\overline{IE}$				14					
G15	$\overline{GAR}$				37					
G16	$\overline{GE}$				36					
G17	LB				8			-0.433		
G18	$\overline{ID}$				35			-2.17		
---	Setup	11,12,14	9,13		Open	30	---	---	---	
G19	S0 Input current, low level	Open	Open		28	Open	0.0	-0.108	mA	
G20	S1 Input current, low level	Open	Open		29	Open	0.0	-0.108	mA	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G21	S2 Input currrent, low level	Open	Open	Open	31	Open	0.0	-0.108	mA	10,9
---	Setup	9,14	11,12,13	Open	Open	30	---	---	---	
G22	M0 Input current, low level	Open	Open	Open	16	Open	0.0	-0.163	mA	
G23	M1				18					
G24	M2				20					
G25	M3				39					
G26	M4				6					
G27	M5				4					
G28	M6				2					
G29	M7				22					

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES.
1. R1 = 1 kΩ ±10%, 1/8 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. When point E is connected apply one clock pulse per figure 9 after all other connections have been made.

*FIGURE 10. Input current (high-level and low-level) test circuit for tables VII and VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE IX. Output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	9,11,12,13,14	Open	30	---	---	---	11,9
H1	GAS Output voltage, high level	↓	↓	26	Open	2.4	4.5	V	↓
H2	RD Output voltage, high level			34	Open	2.4	4.5	V	
H3	IR Output voltage, high level			32	Open	2.4	4.55	V	
---	Setup	9,12	11,13,14	Open	30	---	---	---	
H4	SV Output voltage, high level	9,12	11,13,14	27	Open	2.4	4.5	V	
---	Setup	Open	9,11,12,13,14	Open	30	---	---	---	
H5	GS Output voltage, high level	Open	9,11,12,13,14	38	Open	2.4	4.5	V	
---	Setup	3,5,7,9,12,15,17,19,40	11,13,14,21	Open	30	---	---	---	
H6	PD Output voltage, high level	↓	↓	33	Open	2.4	4.5	V	
H7	VO			23	↓	↓	↓	↓	
H8	V1			24					
H9	V2	↓	↓	25					
---	Setup	2,4,6,11,12,13,16,18,20,22,39	9,14	Open	30	---	---	---	
H10	MO Output voltage, high level	9,11,12	13,14	16	Open	2.4	4.5	V	
H11	M1	↓	↓	18	↓	↓	↓	↓	
H12	M2			20					
H13	M3			39					
H14	M4			6					
H15	M5			4					
H16	M6			2					
H17	M7	↓	↓	22	↓	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE IX. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	9,13,28,29,31	11,12,14	Open	30	---	---	---	11,9
H18	S0 Output voltage, high level	11,12	9,13,14	28	Open	2.4	4.5	V	↓
H19	S1 Output voltage, high level	11,12	9,13,14	29	Open	2.4	4.5	V	
H20	S2 Output voltage, high level	11,12	9,13,14	31	Open	2.4	4.5	V	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage (low-level) test.

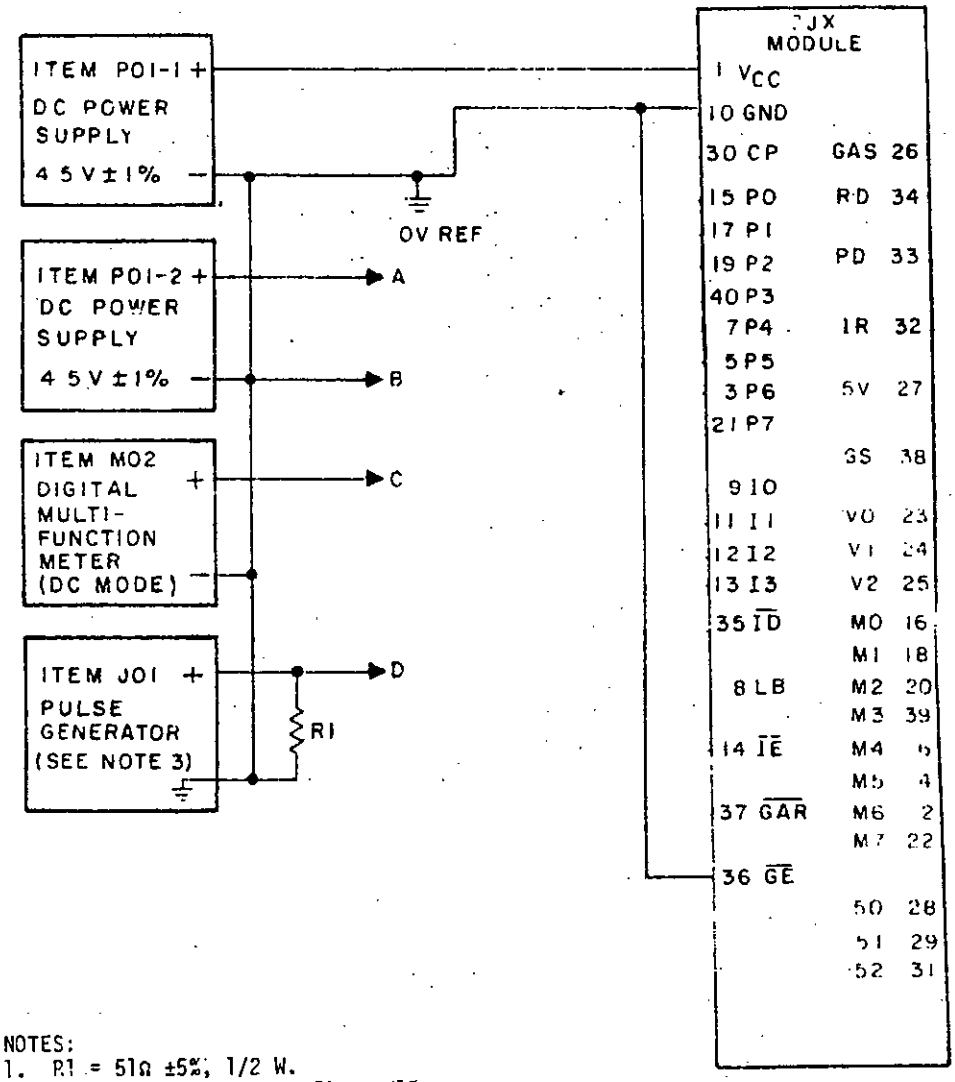
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	9,11,12,13,14	Open	30	---	---	---	11,9
J1	PD Output voltage, low level	Open	9,11,12,13,14	33	Open	0.0	0.5	V	
---	Setup	3,5,7,9,12,17,19, 21,40	11,13,14,15	Open	30	---	---	---	
J2	VO Output voltage, low level			23	Open	0.0	0.5	V	
J3	V1 Output voltage, low level			24	Open	0.0	0.5	V	
J4	V2 Output voltage, low level			25	Open	0.0	0.5	V	
---	Setup	3,5,7,9,12,15,17, 19,40	11,13,14,21	Open	30	---	---	---	
J5	GAS Output voltage, low level			26	Open	0.0	0.5	V	
J6	RD Output voltage, low level			34	Open	0.0	0.5	V	
J7	IR Output voltage, low level			32	Open	0.0	0.5	V	
---	Setup	9,12	11,13,14,21	Open	30	---	---	---	
J8	SV Output voltage, low level	9,12	11,13,14,21	27	Open	0.0	0.5	V	
---	Setup	3,5,7,9,12,15,17, 19,21,40	11,13,14,37	Open	30	---	---	---	
J9	GS Output voltage, low level	3,5,7,9,12,15,17, 19,21,40	11,13,14,37	38	Open	0.0	0.5	V	
---	Setup	11,12,13	2,4,6,9,14,16,18, 20,22,39	Open	30	---	---	---	
J10	MO Output voltage, low level	9,11,12	13,14	16	Open	0.0	0.5	V	
J11	M1			18					
J12	M2			20					
J13	M3			39					
J14	M4			6					
J15	M5			4					
J16	M6			2					
J17	M7			22					

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	9,13	11,12,14,28,29, 31	Open	30	---	---	---	11,9
J18	S0 Output voltage, low level	11,12	9,13,14	28	Open	0.0	0.5	V	
J19	S1 Output voltage, low level	11,12	9,13,14	29	Open	0.0	0.5	V	
J20	S2 Output voltage, low level	11,12	9,13,14	31	Open	0.0	0.5	V	

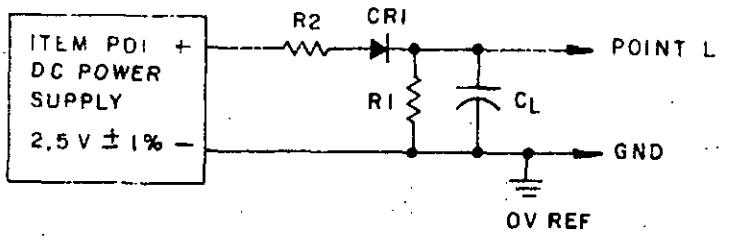
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.



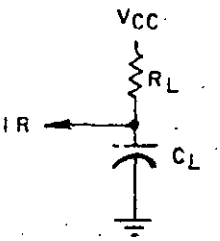
NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs per figure 12.
3. When point D is connected, apply one clock pulse per figure 9, except for setup conditions prior to test conditions H4 and J8, apply two clock pulses to the module.

FIGURE 11. Output voltage (high-level and low-level) test circuit for tables IX and X.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:35Z
Check the source to verify that this is the current version before use.



LOAD CIRCUIT A. (For all outputs except IR)



LOAD CIRCUIT B. (For output IR)

NOTES:
1. Parameters for load circuit A:
   CR1 = 1N4150 or equivalent.
   C_L = 50 pF ±5% (including probe and parasitic capacitance for test conditions P, Q, R, S, T, U, and V).
   R1 = 2.43 kΩ ±1%, 1.8 W.
   R2 = Select value for a current flow of 12 mA ±5% out of point L with point L held at 0.5 V.
2. Parameters for load circuit B:
   R_L = 470Ω ±1%, 1/8 W.
   C_L = 50 pF ±5% (including probe and parasitic capacitance for test conditions P, Q, R, S, T, U, and V).
3. Capacitor C_L not required for dc test parameters.

FIGURE 12. ZJX logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:35Z
Check the source to verify that this is the currenversion before use.

TABLE XI. High-impedance leakage current (high-level) test.

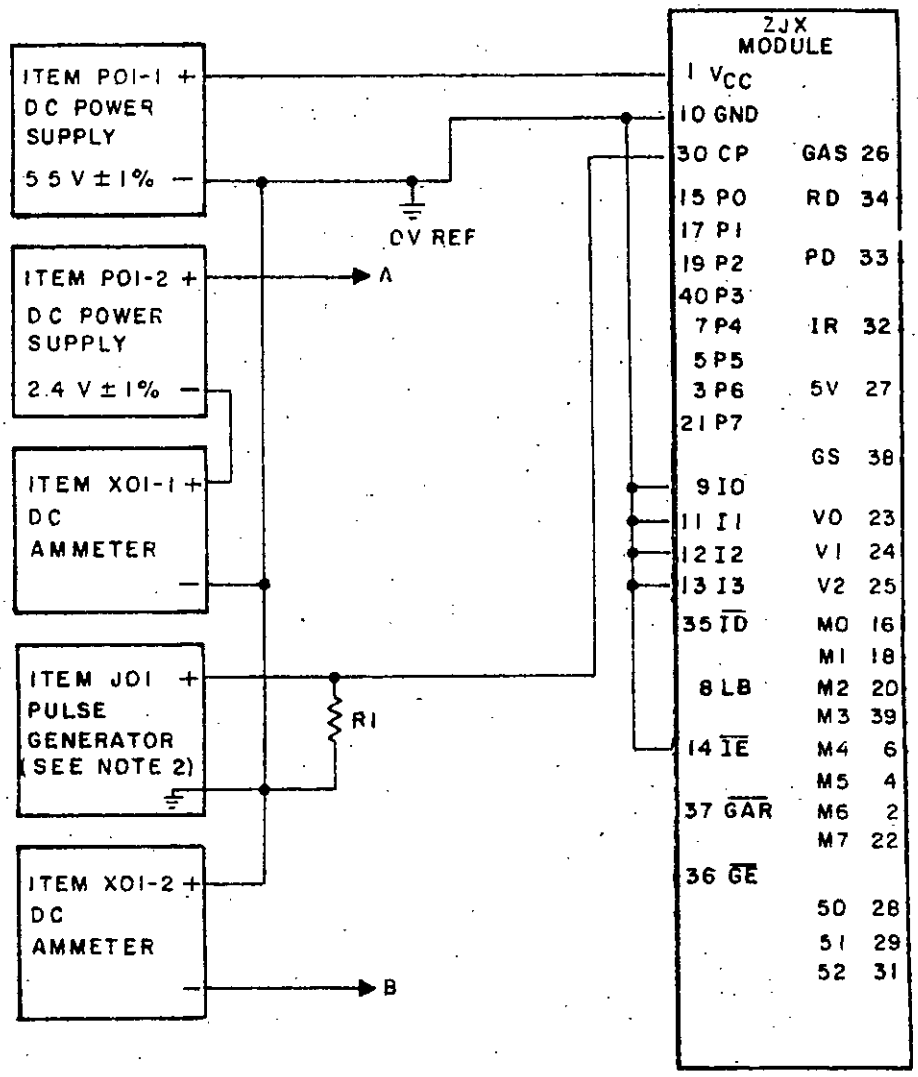
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	V0 High-impedance leakage current, high level	23	Open	-1	50	µA	13,9
K2	V1	24			50		
K3	V2	25			50		
K4	M0	16			150		
K5	M1	18					
K6	M2	20					
K7	M3	39					
K8	M4	6					
K9	M5	4					
K10	M6	2					
K11	M7	22			100		
K12	S0	28			100		
K13	S1	29			100		
K14	S2	31					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

TABLE XII. High-impedance leakage current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	VO High-impedance leakage current, low level	Open	23	1	-50	µA	13,9
L2	V1		24		-50		
L3	V2		25		-50		
L4	M0		16		-150		
L5	M1		18				
L6	M2		20				
L7	M3		39				
L8	M4		6				
L9	M5		4				
L10	M6		2				
L11	M7		22				
L12	S0		28		-100		
L13	S1		29		-100		
L14	S2		31		-100		

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T17:35Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter X01-1 and X01-2 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. After all other connections have been made apply one clock pulse per figure 9, initially, then begin testing.

FIGURE 13. High impedance leakage current (high-level and low-level) test circuit for tables XI and XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

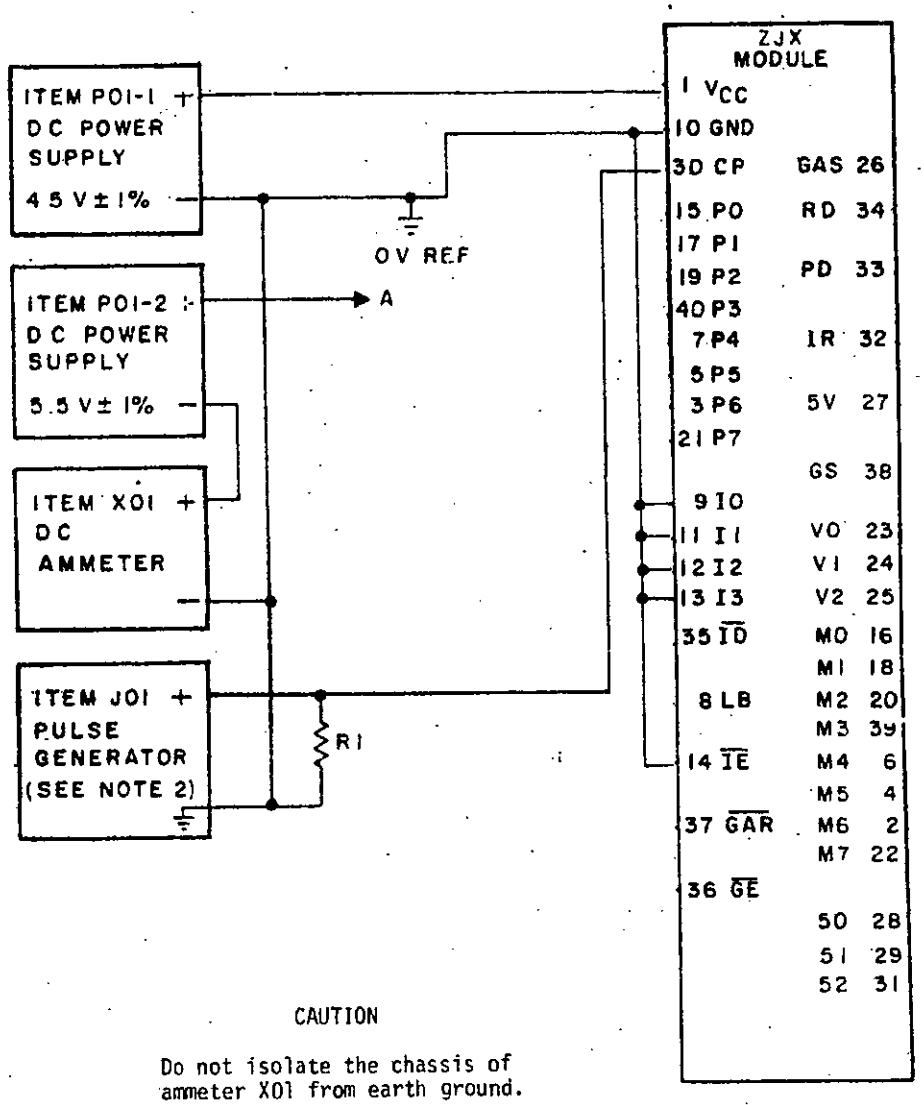
TABLE XIII. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M	IR Output leakage current	32	0	250	µA	14,9

TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N	Power supply current	26	260	mA	15,9

Source: https://assist.dla.mil - Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.



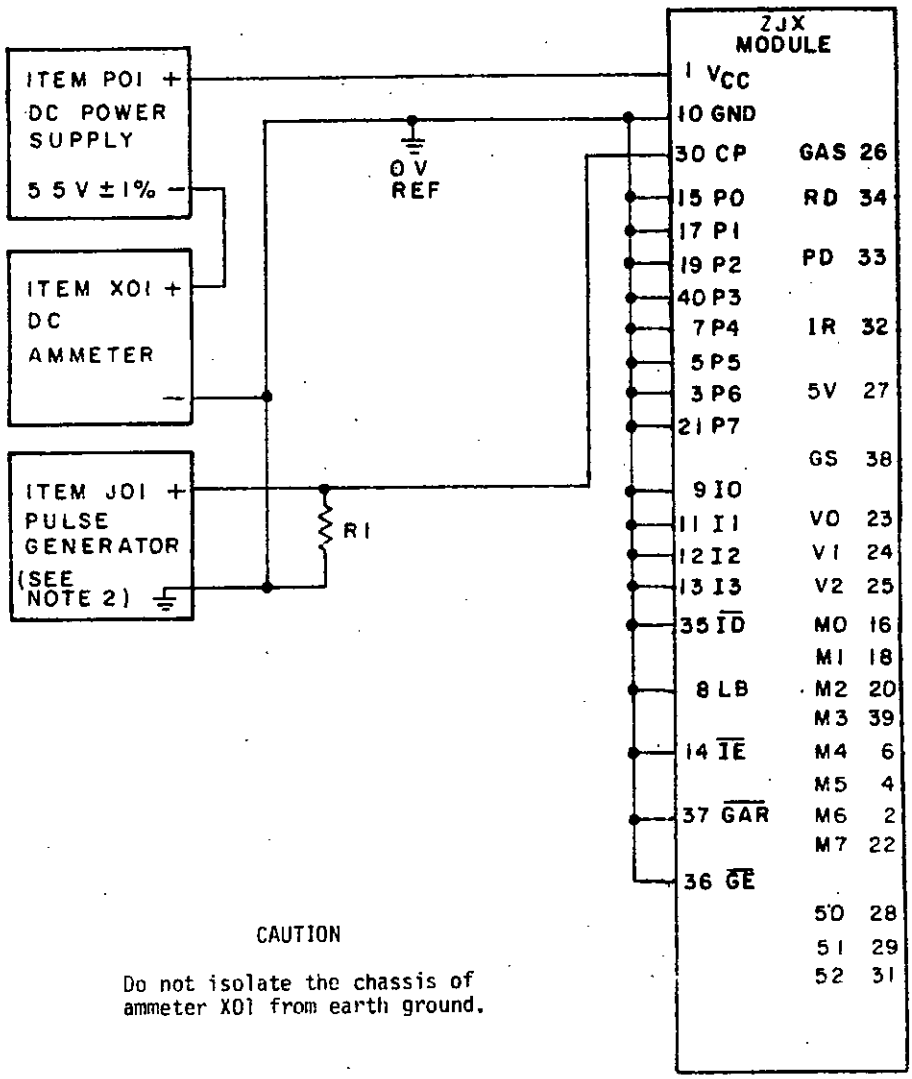
CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. After all other connections have been made apply one pulse per figure 9.

FIGURE 14. Output leakage current test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Place item J01 in single pulse mode and apply one pulse per figure 9.

FIGURE 15. Power supply current test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XV. Dynamic functional test.

Test condition	Parameter	Test description	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P1	Dynamic functional test	Minimum clock pulse low, V_{CC} = 4.5 V	1/, 11/			16
P2		Minimum clock pulse low, V_{CC} = 5.5 V	2/, 11/			16
P3		Minimum clock pulse high, V_{CC} = 4.5 V	3/, 11/			17
P4		Minimum clock pulse high, V_{CC} = 5.5 V	4/, 11/			17
P5		Setup and hold time, minimum cycle time, V_{CC} = 4.5 V	5/ - 11/			18,19, 20,21
P6		Setup and hold time, minimum cycle time, V_{CC} = 5.5 V	5/ - 11/			18,19, 20,21

1/ Test condition P1 checks for minimum clock pulse low time with V_{CC} = 4.5 V. This test also checks setup time-to-clock on SO-S2, MO-M7, GE, GAR, IE, IO-I3, and ID and checks hold time-to-clock on SO-S2, MO-M7, GE, and GAR. Pattern PAT1 checks low-to-high transition setup times and high-to-low transition hold times. Pattern PAT2 checks high-to-low setup times and low-to-high hold times.
2/ Test condition P2 is the same as P1 except V_{CC} = 5.5 V.
3/ Test condition P3 is the same as P1 except minimum clock pulse high time is checked.
4/ Test condition P4 is the same as P3 except V_{CC} = 5.5 V.
5/ Test condition P5, part 1, checks setup time-to-clock on inputs PO-P7 and hold time-to-clock on inputs PO-P7, IE, IO-I3, and ID with V_{CC} = 4.5 V. Test condition P6, part 1, is the same except V_{CC} = 5.5 V. Pattern PAT3 is used for both.
6/ Test condition P5, part 2, checks minimum clock period with V_{CC} = 4.5 V. Test condition P6, part 2, is the same except V_{CC} = 5.5 V. Pattern PAT3 is used for both.
7/ Test condition P5, part 3, checks for LB setup and hold time-to-clock with V_{CC} = 4.5 V. Test condition P6, part 3, is the same except V_{CC} = 5.5 V. The pattern in footnote 10 is used for both.
8/ Test condition P5, part 4, checks for hold time relative to LB on inputs PO-P7 and minimum pulse width-to-accept on PO-P7 with V_{CC} = 4.5 V. Test condition P6, part 4, is the same except V_{CC} = 5.5 V. Pattern A4 is used for both.
9/ Test condition P5, part 5, checks for maximum pulse width-to-reject on inputs PO-P7 with V_{CC} = 4.5 V. Test condition P6, part 5, is the same except V_{CC} = 5.5 V. Pattern A5 is used for both.
10/ Pattern for P5 and P6, part 3. L = 0.5 +0.1, -0.0 V, H = 2.4 +0.0, -0.1 V, GE and IE = L always; ID and GAR = H always.

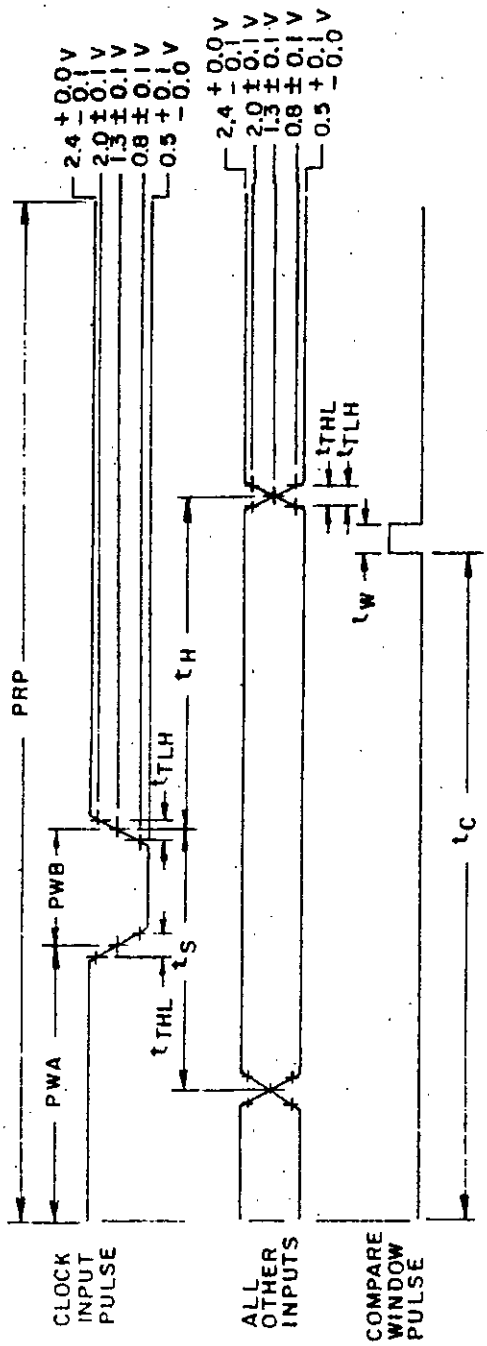
I3	I2	I1	I0	P7	P6	P5	P4	P3	P2	P1	P0	IR
L	L	L	H	H	H	H	H	H	H	H	H	H
L	H	L	H	L	L	L	L	L	L	L	L	H

11/ Test condition P1-P6 apply the functional test vectors (see 6.6). Each test condition tests rainly for the test parameters as described in the test description, but also for the following operating conditions and parameters:

Loaded output voltage levels (loaded per figure 12):
V_{OH} = 2.4 V minimum.
V_{OL} = 0.5 V maximum.

Worst case input threshold voltage levels:
V_{IH} = 2.4 V minimum.
V_{IL} = 0.5 V maximum.

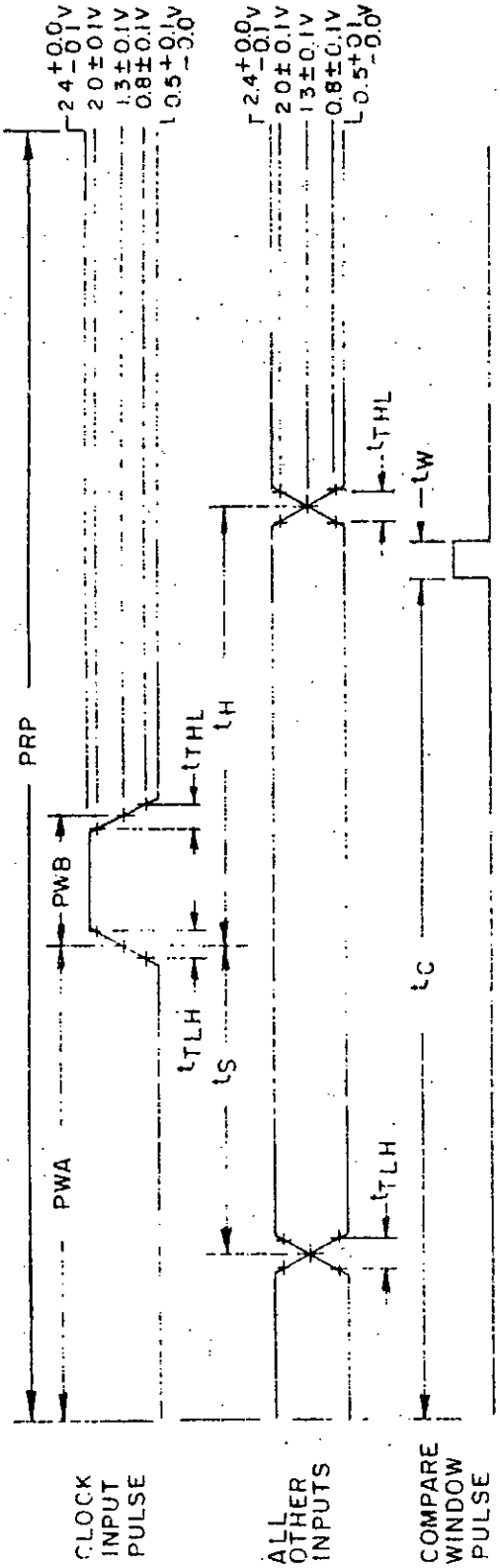
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 90 ±2 ns.
   PWB = 31 ±1 ns.
   PRP = 328 ±2 ns.
   t_C = 300 ±5 ns.
   t_W = 10 ±2 ns.
   $t_{TLH} = t_{THL}$ = 5 ±5 ns.
   t_S = 16 ±1 ns (S0-S2, M0-M7, $\overline{GE}$, $\overline{GAR}$).
   = 71 ±1 ns (I0-I3, $\overline{TE}$).
   = 43 ±1 ns ($\overline{ID}$).
   t_H = 11 ±1 ns (S0-S2, M0-M7).
   = 14 ±1 ns ($\overline{GE}$, $\overline{GAR}$).
   = 308 ±2 ns (I0-I3, $\overline{TE}$).
   = 280 ±2 ns ($\overline{ID}$).
2. Other inputs are held at a constant level for the entire cycle, high level = 2.4 +0.0, -0.1 V, low level = 0.5 +0.1, -0.0 V.

FIGURE 16. Dynamic functional test waveforms (minimum clock pulse low) for table XV.

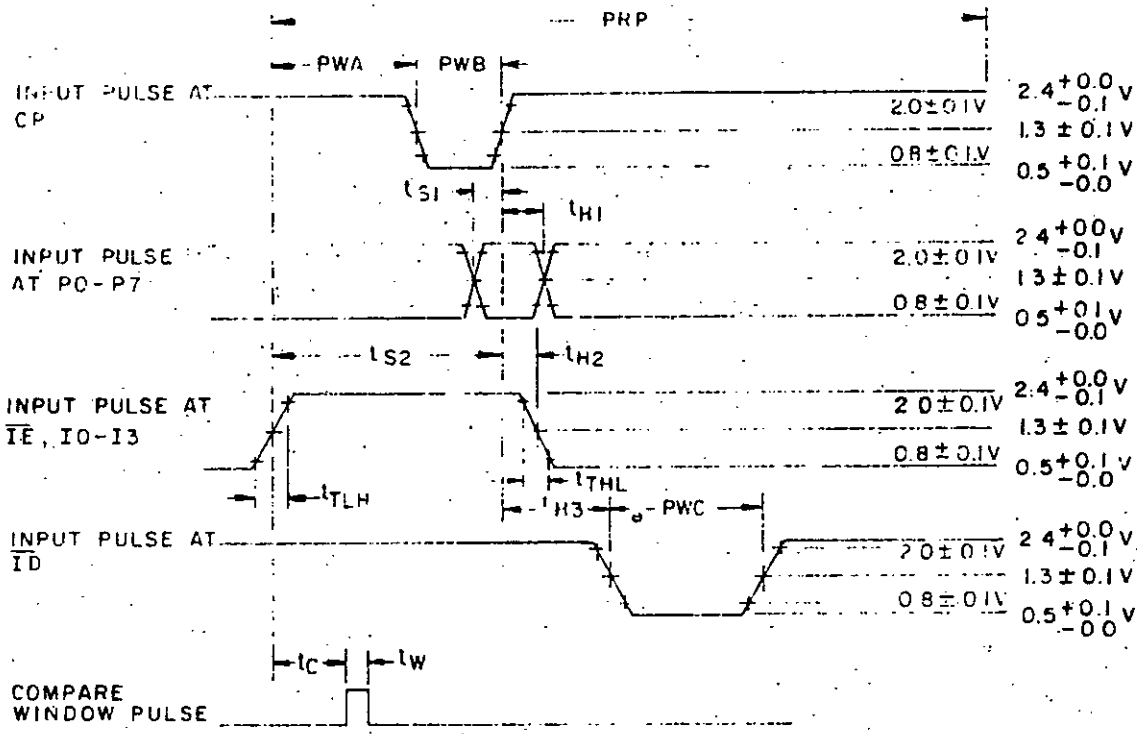
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 90 ±2 ns.
   PWB = 31 ±1 ns.
   PRP = 328 ±2 ns.
   $t_{TLH} = t_{THL}$ = 5 ±5 ns.
   tC = 300 ±5 ns.
   tW = 10 ±2 ns.
   tS = 90 ±2 ns.
   tH = 238 ±2 ns.
2. Other inputs are held at a constant level for the entire cycle, high level = 2.4 +0.0, -0.1 V, low level = 0.5 +0.1, -0.0 V.

FIGURE 17. Dynamic functional test waveforms (minimum clock pulse high) for Table XV.

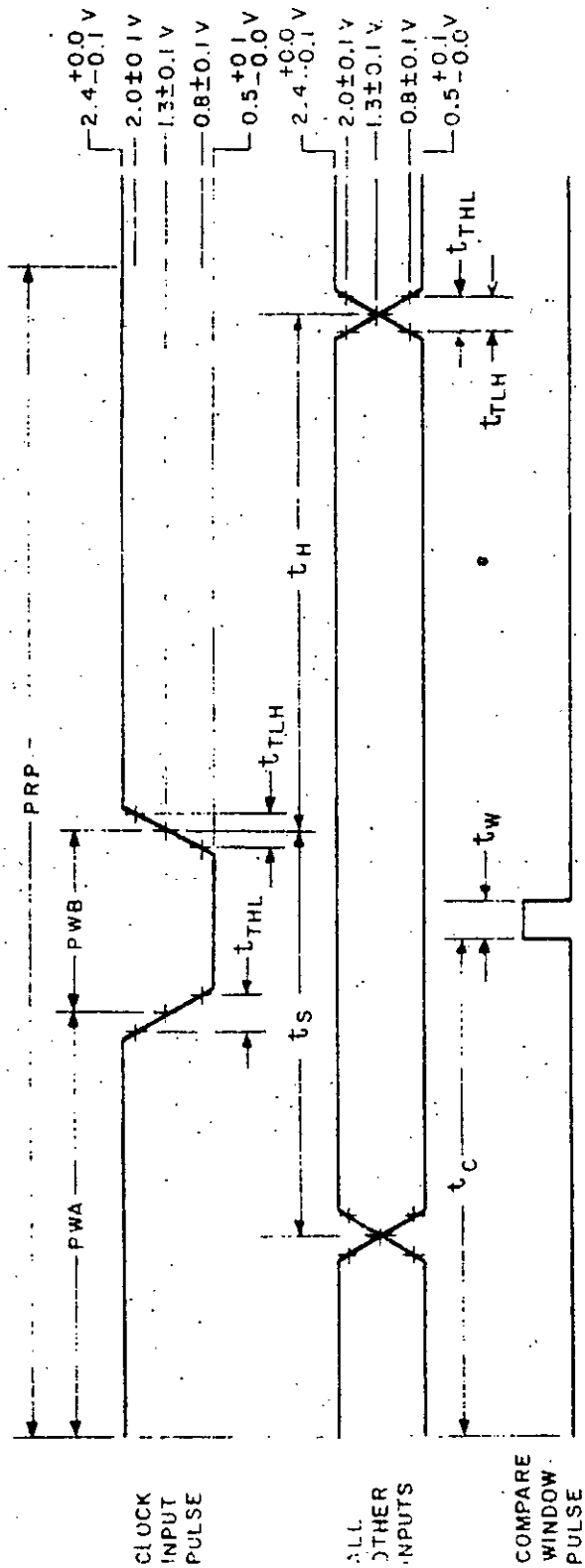
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 200 +1 ns.
   PWB = 100 ±1 ns.
   PWC = 200 ±1 ns.
   PRP = 500 ±5 ns.
   t_{S1} = 16 ±1 ns.
   t_{S2} = 300 ±1 ns.
   t_{H1} = 9 ±1 ns.
   t_{H2} = 1 ±1 ns.
   t_{H3} = 1 ±1 ns.
   t_C = 100 ±1 ns.
   t_W = 10 ±2 ns.
   t_{TLH} = t_{THL} = 5 ±5 ns.
2. Other inputs are held at a constant level for the entire cycle, high level = 2.4 +0.0, -0.1 V, low level = 0.5 +0.1, -0.0 V.

* FIGURE 18. Dynamic functional test waveforms (setup and hold time) for table XV.

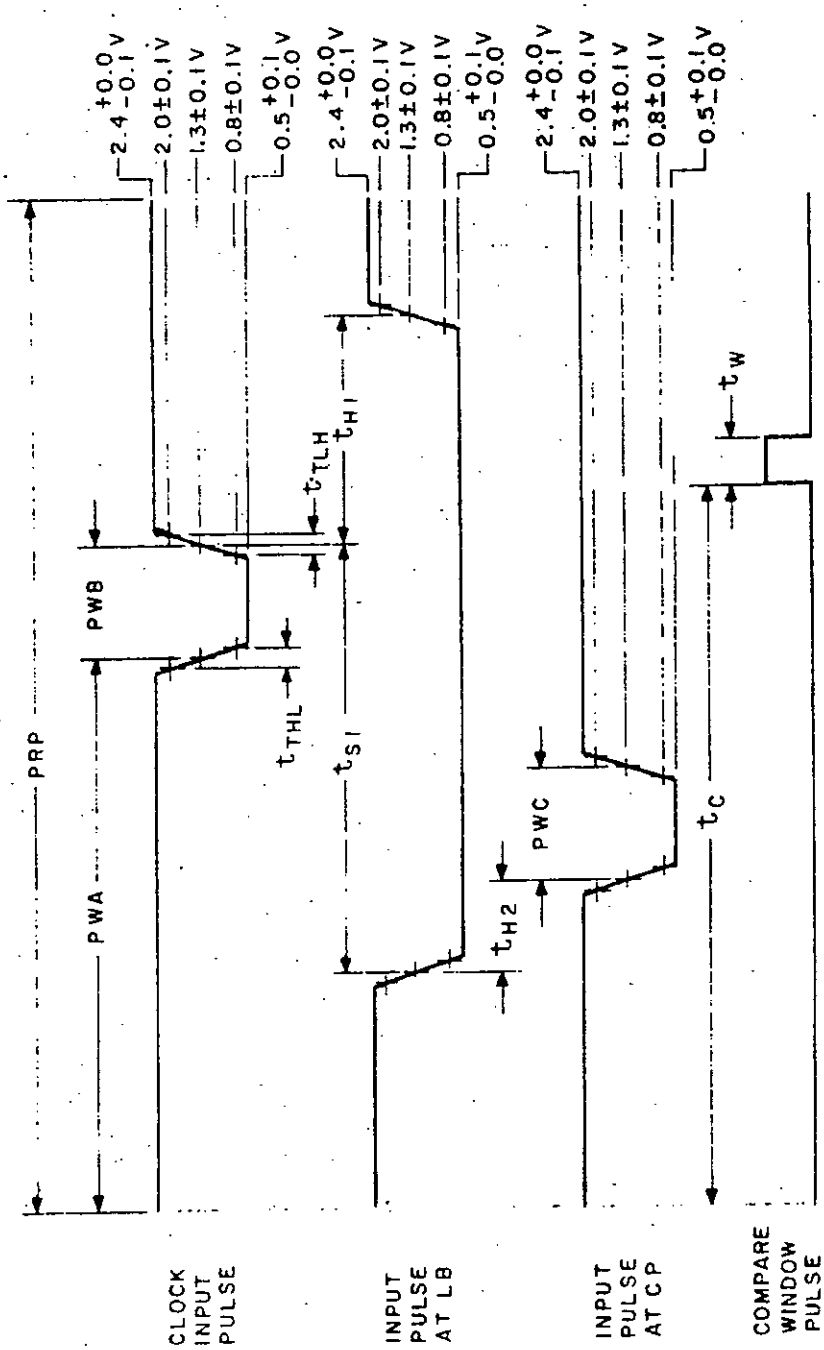
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 40 ±1 ns.
   PWB = 31 ±1 ns.
   PRP = 111 ±1 ns.
   tC = 35 ±1 ns.
   tW = 8 ±1 ns.
   tS = 70 ±1 ns.
   tH = 40 ±1 ns.
   tTLH = tTHL = 5±5 ns.
2. Other inputs are held constant for the entire cycle, high level = 2.4 +0.0, -0.1 V, low level = 0.5 +0.1, -0.0 V.

* FIGURE 19. Dynamic functional test waveforms (setup and hold time) for table XV.

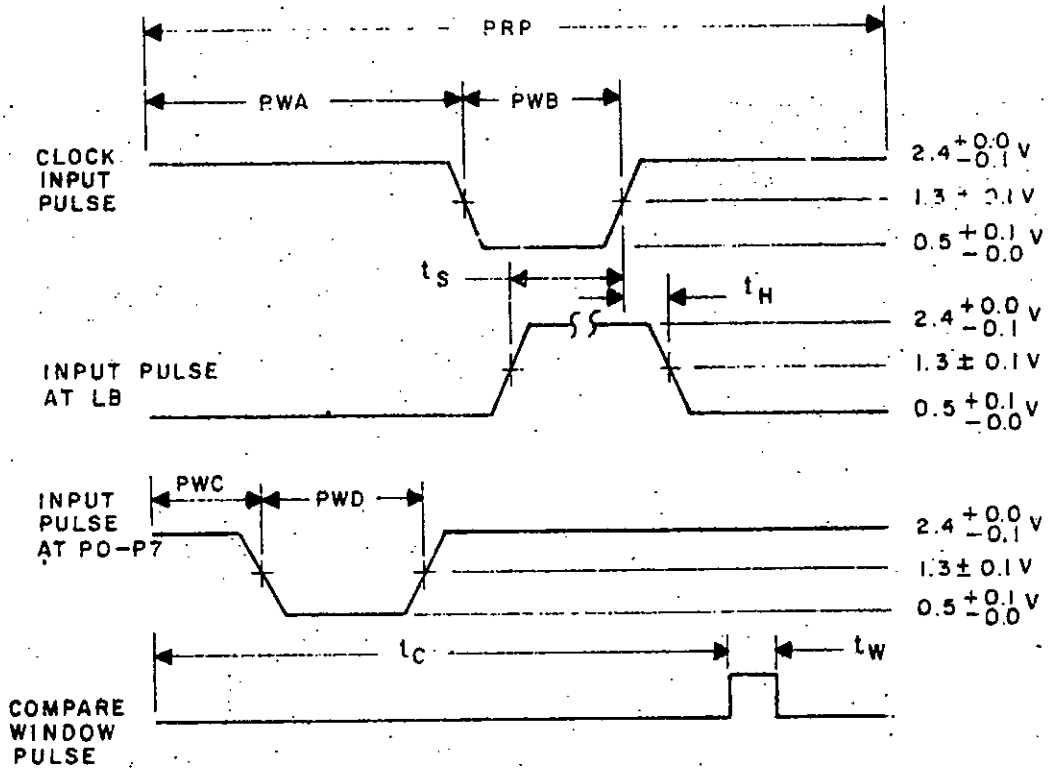
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 400 ±1 ns.
   PWB = 100 ±1 ns.
   PWC = 41 +T ns for part 4.
   = 7 +T ns for part 5.
   PRP = 1,000 ±1 ns.
   t_{S1} = 300 ±1 ns.
   t_{H1} = 400 ±1 ns.
   t_{H2} = 26 +T ns for part 4.
   = 40 ±1 ns for part 5.
   t_C = 650 ±1 ns.
   t_W = 10 ±7 ns.
   t_{TLH} = t_{THL} = 5±5 ns.
2. Other inputs are held constant for the entire cycle, high level = 2.4 +0.0, -0.1 V, low level = 0.5 +0.1, -0.0 V.

* FIGURE 20. Dynamic functional test waveforms (setup and hold time) for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 400 ±1 ns.
   PWB = PWC = PWD = 100 ±1 ns.
   PRP = 1,000 ±1 ns.
   t_S = 481 ±1 ns.
   t_H = 1 +1 ns.
   t_C = 800 ±1 ns.
   t_W = 10 ±2 ns.
   $t_{TLH} = t_{THL}$ = 5 ±5 ns (all inputs) measured at the 0.8 ±0.1 V and 2.0 ±0.1 V levels.
2. Other inputs are held constant for the entire cycle, high level = 2.4 ±0.0, -0.1 V, low level = 0.5 +0.1, -0.0 V.

* FIGURE 21. Dynamic functional test waveforms (setup and hold time) for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
---	Setup	9,11,12, 13,14	Open	Open	Open	Open	Open	---	---	---	22,23
Q1	PD Transition time (t_{TLH}), C_L = 50 pF	11,13,14, 36	9,12	21			33	1	17	ns	
Q2	RD	11,13,14, 36	9,12	21			34	1	17	ns	
Q3	GAS	11,13,14, 36	9,12	21			26	1	17	ns	
---	Setup	9,11,12, 13,14	Open	Open			Open	---	---	---	
Q4	GS	11,12,14	9,13	36			38	1	17	ns	
Q5	SY	11,13,14	9,12	Open	21	30	27	1	17	ns	

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XVII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
---	Setup	9,11,12, 13,14	Open	Open	Open	Open	Open	---	---	---	22,23
R1	IR Transition time (t_{THL}), C_L = 50 pF	11,13,14, 36	9,12	21			32	1	6	ns	
R2	PD						33				
R3	RD						34				
R4	GAS						26				
---	Setup	9,11,12, 13,14	Open	Open			Open	---	---	---	
R5	GS Transition time (t_{THL}), C_L = 50 pF	11,12,14	9,13	36			38	1	6	ns	
R6	SV Transition time (t_{THL}), C_L = 50 pF	11,13,14	9,12	Open	21	30	27	1	6	ns	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XVIII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
---	Setup	9,11,12, 13,14	Open	Open	Open	Open	Open	---	---	---	22,23
S1	$\overline{IE}$ to GAS Propagation delay time (t_{PLH}), C_L = 50 pF	11,13,21, 36,37	9,12	14		14	26	4	47	ns	
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
---	Setup	11,13,14, 36,37,21	9,12	Open		Open	Open	---	---	---	
S2	I0-I3 to GAS Propagation delay time (t_{PLH}), C_L = 50 pF	11,13,14, 36,37	Open	9,12		9	26	4	47	ns	
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
S3	$\overline{ID}$ to IR Propagation delay time (t_{PLH}), C_L = 50 pF	11,13,14, 17,36	9,12	35		35	32	4	42	ns	
S4	$\overline{ID}$ to RD	9,11,12, 13,14,36	Open	35		35	34	2	13	ns	
S5	ID to GAS Propagation delay time (t_{PLH}), C_L = 50 pF	11,13,14, 21,36	9,12	35		35	26	2	23	ns	
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
S6	P7 to IR Propagation delay time (t_{PLH}), C_L = 50 pF	11,13,14 36	9,12	21		30	32	8	82	ns	
S7	P7 to PD						33	5	57		
S8	P7 to RD						34	5	57		
S9	P7 to GAS						26	6	66		
S10	$\overline{GE}$ to PD	11,12,14	9,13	36			33	4	42		
S11	$\overline{GE}$ to RD	11,12,14	9,13	36			34	4	45		
S12	$\overline{GE}$ to GS	11,12,14	9,13	36			38	3	32		
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
S13	I1 to IR Propagation delay time (t_{PLH}), C_L = 50 pF	14,15	9,12, 13	11		30	32	5	56	ns	

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
S14	[illegible] to SV Propagation delay time (t_{PLH}), C_L = 50 pF	11,13,14	9,12	Open	21	30	27	2	25	ns	22,23
---	Setup	9,11,12, 13,14	Open	Open	Open	Open	Open	---	---	---	
S15	[illegible] to IR Propagation delay time (t_{PLH}), C_L = 50 pF	9,14,21, 36	11,12, 13	22		30	32	8	82	ns	
S16	[illegible] to PD Propagation delay time (t_{PLH}), C_L = 50 pF	9,14,21, 36	11,12, 13	22		30	33	5	57	ns	
S17	[illegible] to RD Propagation delay time (t_{PLH}), C_L = 50 pF	9,14,21, 36	11,12, 13	22		30	34	5	57	ns	
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
S18	S0-S2 to IR Propagation delay time (t_{PLH}), C_L = 50 pF	11,12,14, 15,36	9,13	28,29, 31		30	32	7	74	ns	

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XIX. Propagation delay time (t_{PHL}) test.

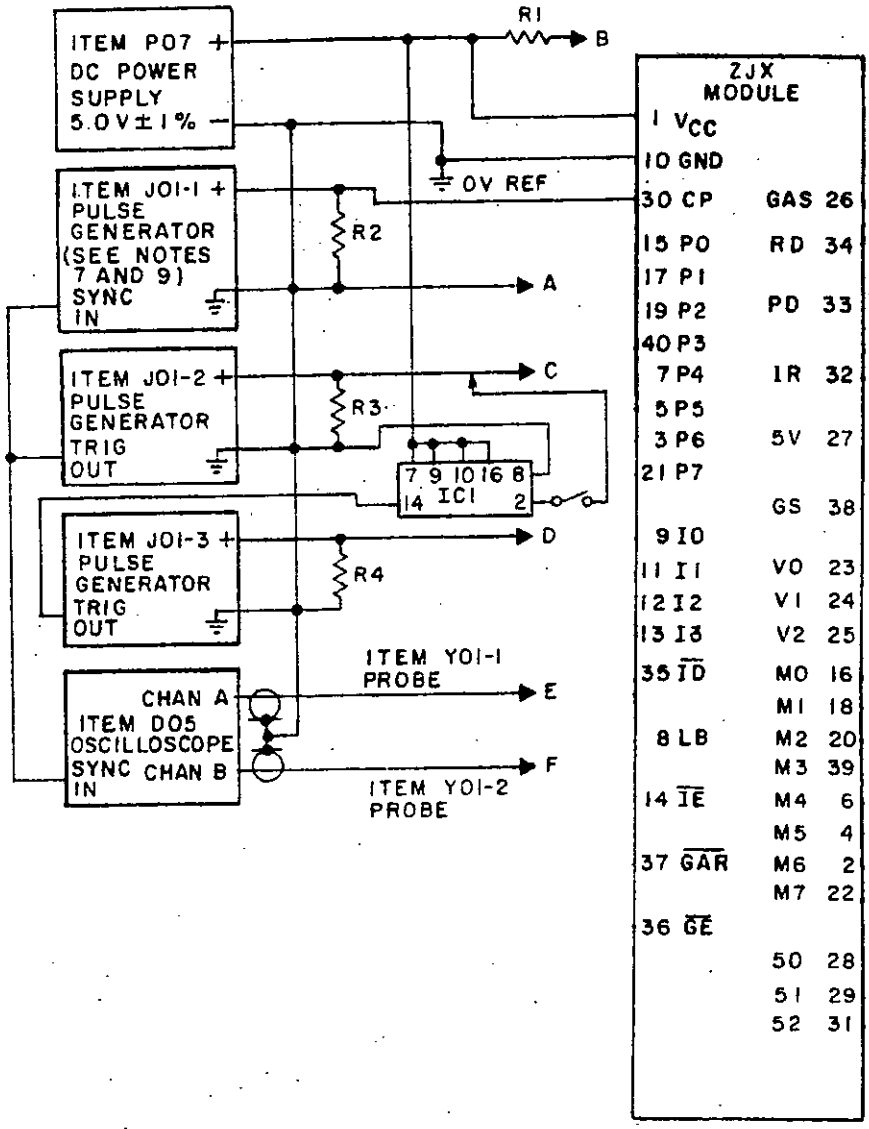
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
---	Setup	9,11,12, 13,14	Open	Open	Open	Open	Open	---	---	---	22,23
T1	$\overline{IE}$ to GAS Propagation delay time (t_{PHL}), C_L = 50 pF	11,13,21, 36,37	9,12	14		14	26	4	47	ns	
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
---	Setup	11,13,14, 36,37,21	9,12	Open		Open	Open	---	---	---	
T2	I0-I3 to GAS Propagation delay time (t_{PHL}), C_L = 50 pF	11,13,14, 36,37	Open	9,12		9	26	4	47	ns	
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
T3	$\overline{ID}$ to IR Propagation delay time (t_{PHL}), C_L = 50 pF	11,13,14, 17,36	9,12	35		35	32	4	42	ns	
T4	$\overline{ID}$ to RD	9,11,12, 13,14,36	Open	35		35	34	2	13	ns	
T5	$\overline{ID}$ to GAS Propagation delay time (t_{PHL}), C_L = 50 pF	11,13,14, 21,36	9,12	35		35	26	2	23	ns	
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
T6	P7 to IR Propagation delay time (t_{PHL}), C_L = 50 pF	11,13,14 36	9,12	21		30	32	8	82	ns	
T7	P7 to PD						33	5	57		
T8	P7 to RD						34	5	57		
T9	P7 to GAS						26	6	66		
T10	$\overline{GE}$ to PD	11,12,14	9,13	36			33	4	42		
T11	$\overline{GE}$ to RD	11,12,14,	9,13	36			34	4	45		
T12	$\overline{GE}$ to GS	11,12,14	9,13	36			38	3	32		

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
---	Setup	9,11,12, 13,14	Open	Open	Open	Open	Open	---	---	---	22,23
T13	I1 to IR Propagation delay time (t_{PHL}), C_L = 50 pF	14,15	9,12, 13	11	Open	30	32	5	56	ns	
T14	P7 to SV Propagation delay time (t_{PHL}), C_L = 50 pF	11,13,14	9,12	Open	21	30	27	2	30	ns	
---	Setup	9,11,12, 13,14	Open	Open	Open	Open	Open	---	---	---	
T15	M7 to IR Propagation delay time (t_{PHL}), C_L = 50 pF	9,14,21, 36	11,12, 13	22		30	32	8	82	ns	
T16	M7 to PD Propagation delay time (t_{PHL}), C_L = 50 pF	9,14,21, 36	11,12, 13	22		30	33	5	57	ns	
T17	M7 to RD Propagation delay time (t_{PHL}), C_L = 50 pF	9,14,21, 36	11,12, 13	22		30	34	5	57	ns	
---	Setup	9,11,12, 13,14	Open	Open		Open	Open	---	---	---	
T18	S0-S2 to IR Propagation delay time (t_{PHL}), C_L = 50 pF	11,12,14, 15,36	9,13	28,29, 31		30	32	7	74	ns	

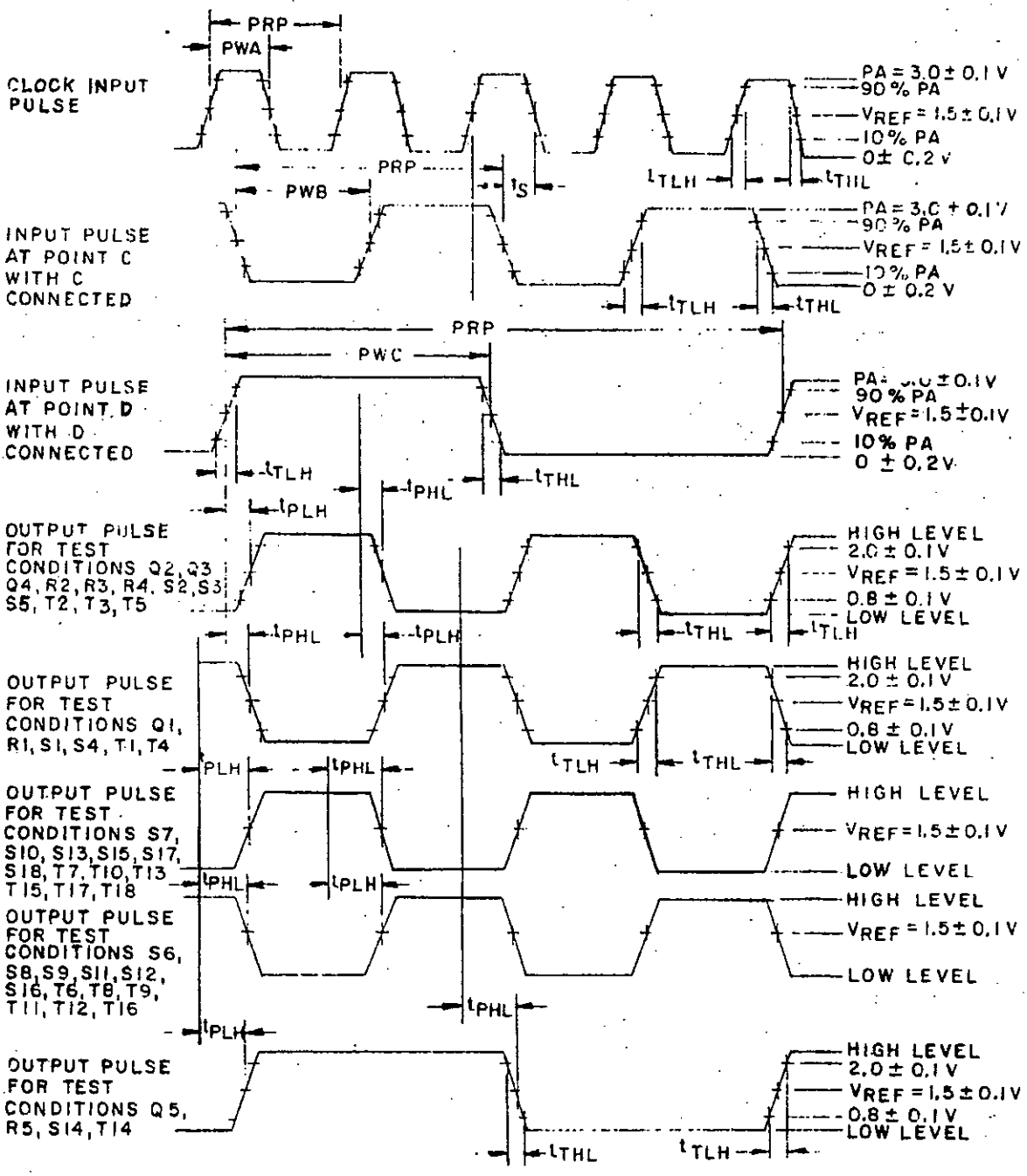
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1.0 kΩ ±10%, 1/8 W.
2. R2-R4 = 51Ω ±5%, 1/2 W.
3. IC1 = 9316 or equivalent.
4. S1 = SPST switch.
5. Close S1 for test conditions Q5, R6, S14 and T14.
6. Load all outputs per figure 12.
7. When setup is called for, place item J01-1 in single pulse mode and pulse the module one time.
8. The clock pulse is not used for test conditions S2 and T2.
9. Sequence for test conditions S3 and T3: After indicated setup conditions leave item J01-1 in the single pulse mode and pulse the module once, then take CP high and make the measurement.

* FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI through XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters:
   Clock input:
   PWA = 250 +25 ns.
   PRP = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

   Point C:
   PWB = 500 +50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   t_S = 125 ±12 ns.

   Point D:
   PWC = 1,000 ±100 ns.
   PRP = 2,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the measurements, verify that the low level is ≤ 0.5 V and that the high level is ≥ 2.4 V.

* FIGURE 23. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI through XIX.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XX. Output crosstalk (low-level) test.

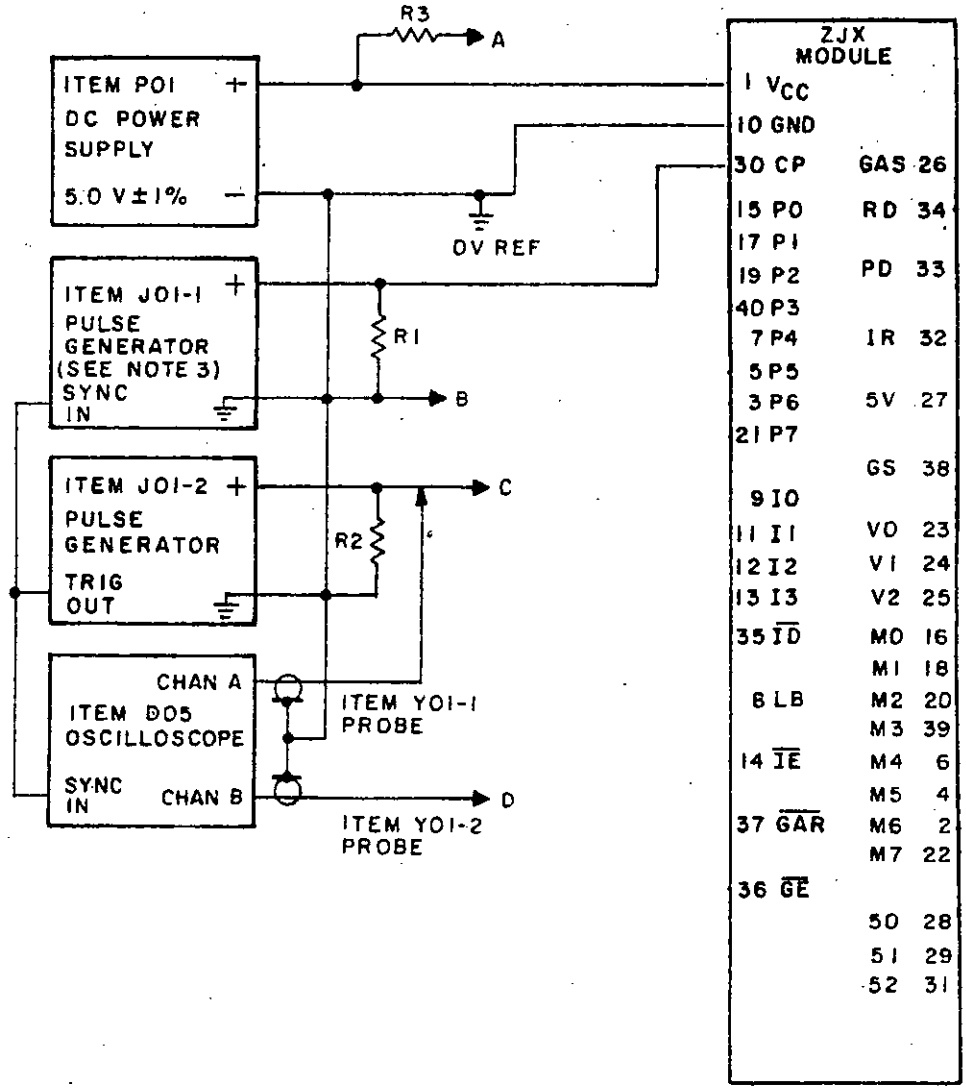
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	9,11,12,13, 14	Open	Open	---	---	---	24,25
U1	RD Output crosstalk, low level	9,12	11,13,14,15, 36,37	21	34	0	1,000	mV	
U2	IR Output crosstalk, low level		11,13,14,15, 36,37	21	32	0	1,000	mV	
U3	PD Output crosstalk, low level		11,13,14	35	33	0	1,000	mV	
---	Setup		11,13,14, 21,36	Open	Open	---	---	---	
U4	SV Output crosstalk, low level	11,12	9,13,14,36	21	27	0	1,000	mV	
---	Setup	Open	9,11,12,13, 14	Open	Open	---	---	---	
U5	GS Output crosstalk, low level	Open	13,14,36,37	9,11,12	38	0	1,000	mV	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXI. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	9,11,12,13, 14	Open	Open	---	---	---	24,25
V1	GS Output crosstalk, high level	9,12	11,13,14, 36	21	38	0	700	mV	
V2	PD		11,13,14,15, 36,37		33				
V3	VO		11,13,14, 17,36,37		23				
V4	V1		11,13,14,19, 36,37		24				
V5	V2		11,13,14,7, 36,37		25		900		
---	Setup	9,12,13	11,14,16	Open	Open	---	---	---	
V6	IR Output crosstalk, high level	9,12	11,13,14,37	21	32	0	700	mV	
---	Setup	9,12,21	11,13,14,36	Open	Open	---	---	---	
V7	SV Output crosstalk, high level	9,12	11,13,14, 36,37	35	27	0	700	mV	
---	Setup	Open	9,11,12,13, 14	Open	Open	---	---	---	
V8	RD Output crosstalk, high level	Open	13,14	9,11,12	34	0	700	mV	
V9	GAS Output crosstalk, high level	Open	13,14	9,11,12	26	0	700	mV	

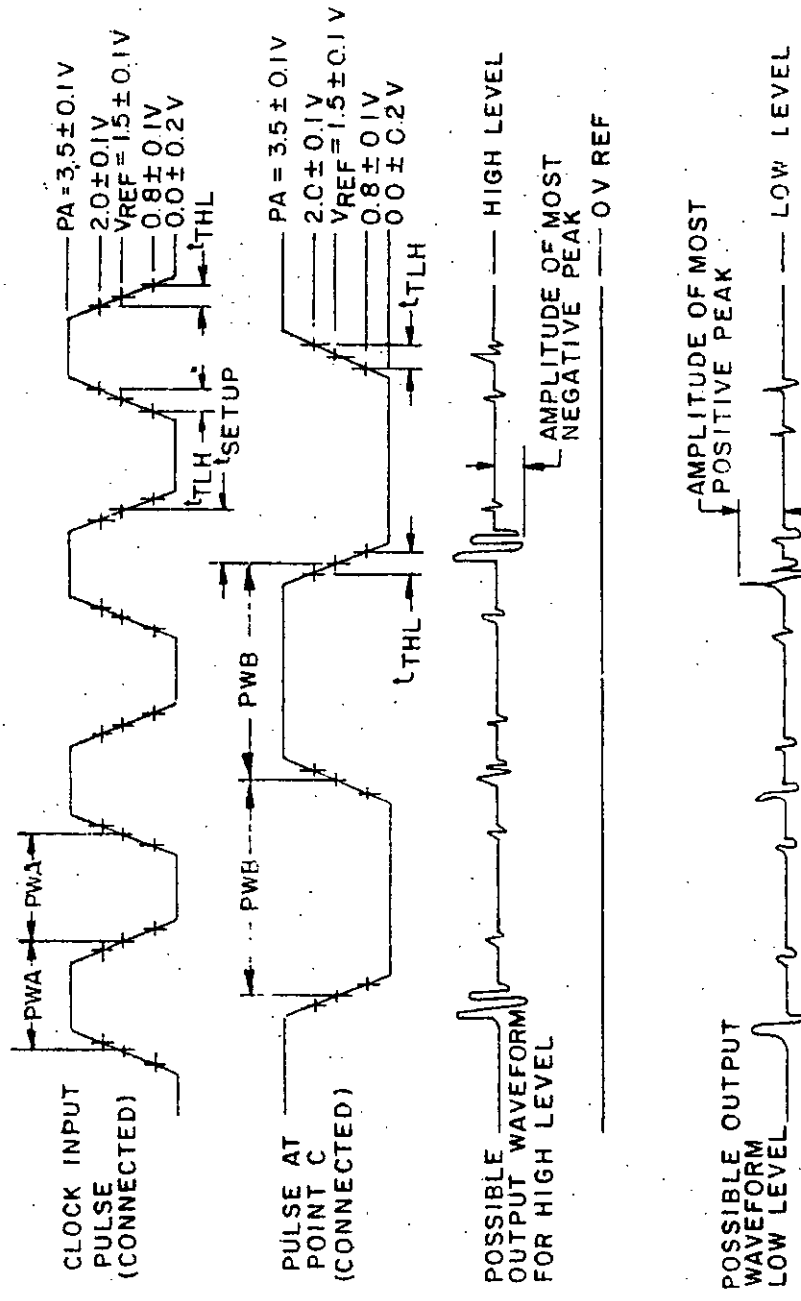
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 1 kΩ ±1%, 1/8 W.
3. During changes to instruction codes I0-I3, clock must be high.
4. Load all outputs per figure 12, including M0-M7 and S0-S2.

* FIGURE 24. Output crosstalk (low-level and high-level) test circuit for tables XX and XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES: Input pulse parameters:
PWA = 250 ±25 ns.
PWB = 500 ±50 ns.
t_{SETUP} = 125 +12 ns.
$t_{TLH} = t_{THL}$ = 10 ±5 ns.

* FIGURE 25. Output crosstalk (low-level and high-level) test circuit waveforms for tables XX and XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXII. Power supply transient amplitude test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
---	Setup	9,11,12,13, 14	Open	Open	---	---	---	26,27
W	Power supply transient amplitude	11,13,14,36	9,12	21	0	400	mV	26,27

* TABLE XXIII. Power supply filter capacitance test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X	Power supply filter capacitance	Open	Open	1	10	3.15	5.85	µF	28

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXIV. Input capacitance test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y1	CP Input capacitance	30	10	Open	Open	2	25	pF	28
Y2	P0	15							
Y3	P1	17							
Y4	P2	19							
Y5	P3	40							
Y6	P4	7							
Y7	P5	5							
Y8	P6	3							
Y9	P7	21							
Y10	I0	9							
Y11	I1	11							
Y12	I2	12							
Y13	I3	13							
Y14	$\overline{ID}$	35					45		
Y15	LB	8					25		
Y16	$\overline{IE}$	14					25		
Y17	$\overline{GAR}$	37					35		
Y18	$\overline{GE}$	36					35		

Source: https://assist.dla.mil --Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXV. Input/output capacitance test.

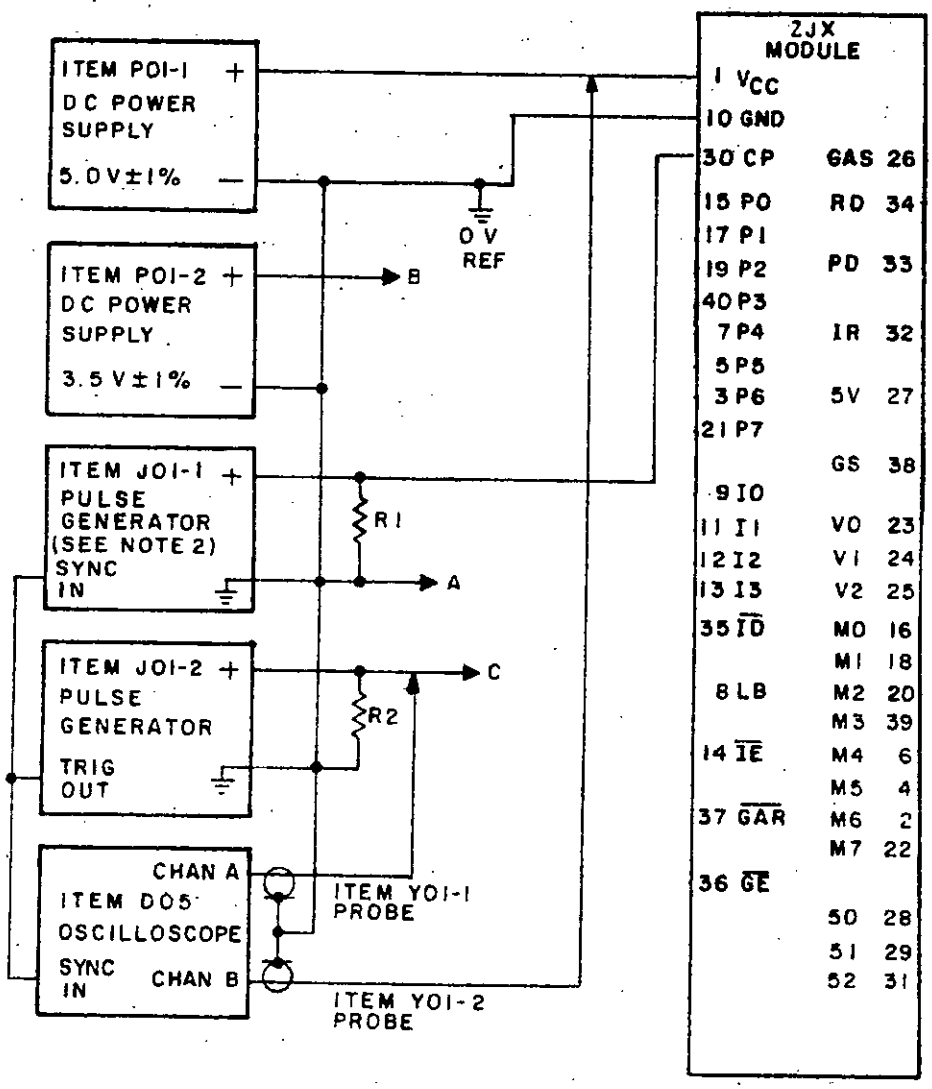
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Z1	M0 Input/output capacitance	16	10	Open	Open	2	50	pF	28
Z2	M1	18							
Z3	M2	20							
Z4	M3	39							
Z5	M4	6							
Z6	M5	4							
Z7	M6	2							
Z8	M7	22					40		
Z9	S0	28					40		
Z10	S1	29					40		
Z11	S2	31							

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXVI. Output capacitance test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA1	GAS Output capacitance	26	10	Open	Open	2	35	pF	28
AA2	RD	34					25		
AA3	PD	33					25		
AA4	IR	32					25		
AA5	SV	27					40		
AA6	GS	38							
AA7	VO	23							
AA8	V1	24							
AA9	V2	25							

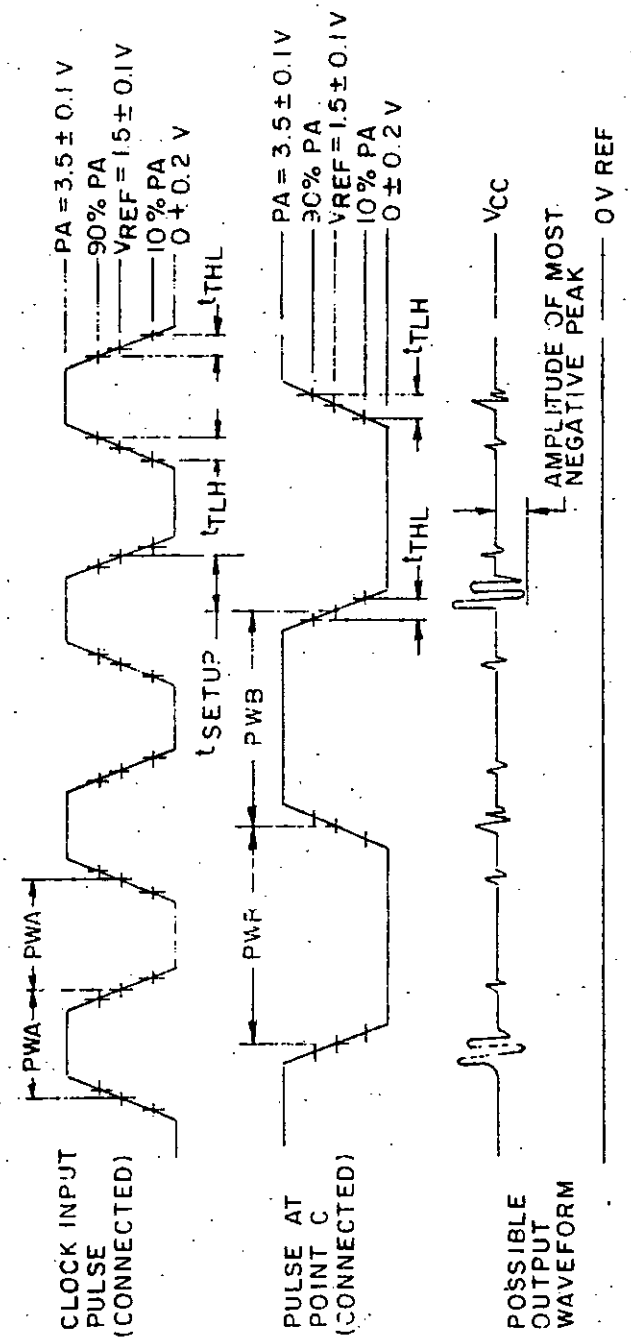
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. During changes to instruction codes I0-I3 clock must be high.

* FIGURE 26. Power supply transient amplitude test circuit for table XXII.

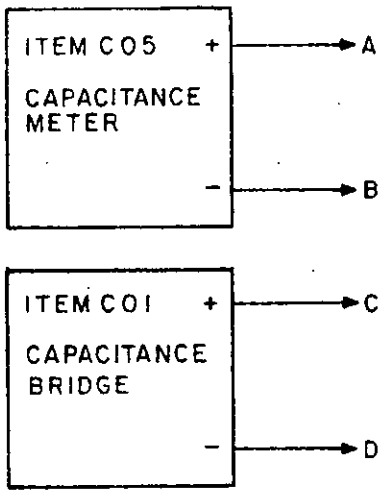
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PWA = 250 ±25 ns.
PWB = 500 ±50 ns.
t_{SETUP} = 125 ±12 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

* FIGURE 27. Power supply transient amplitude test circuit waveforms for table XXII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



ZJX MODULE			
1	V_{CC}		
10	GND		
30	CP	GAS	26
15	P0	RD	34
17	P1		
19	P2	PD	33
40	P3		
7	P4	IR	32
5	P5		
3	P6	SV	27
21	P7		
		GS	38
9	I0		
11	I1	V0	23
12	I2	V1	24
13	I3	V2	25
35	$\overline{ID}$	M0	16
		M1	18
8	LB	M2	20
		M3	39
14	$\overline{IE}$	M4	6
		M5	4
37	$\overline{GAR}$	M6	2
		M7	22
36	$\overline{GE}$		
		S0	28
		S1	29
		S2	31

NOTES:
1. Do not connect test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.
3. Capacitance bridge (C01) setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 28. Power supply filter, input, input/output and output capacitance test circuit for tables XXIII through XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXVII. Disable/enable recovery time test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	9,11,12,13,14	Open	Open	---	---	---	29,30
---	Setup	11,12,13	9,14,16,18,20, 39,6,4,2,22	Open	Open	---	---	---	
---	Setup	9,11,12	13,14	Open	Open	---	---	---	
BB1	IE to M0 (t_{PLZ}) Disable/enable recovery time, C_L = 50 pF		13	14	16	5	52	ns	
BB2	IE to M0 (t_{PZL})				16				
BB3	IE to M1 (t_{PLZ})				18				
BB4	IE to M1 (t_{PZL})				18				
BB5	IE to M2 (t_{PLZ})				20				
BB6	IE to M2 (t_{PZL})				20				
BB7	IE to M3 (t_{PLZ})				39				
BB8	IE to M3 (t_{PZL})				39				
BB9	IE to M4 (t_{PLZ})				6				
BB10	IE to M4 (t_{PZL})				6				
BB11	IE to M5 (t_{PLZ})				4				
BB12	IE to M5 (t_{PZL})				4				
BB13	IE to M6 (t_{PLZ})				2				
BB14	IE to M6 (t_{PZL})				2				
BB15	IE to M7 (t_{PLZ})				22				
BB16	IE to M7 (t_{PZL})				22				
---	Setup	Open	9,11,12,13,14	Open	Open	---	---	---	
---	Setup	11,12,13,16,18, 20,39,6,4,2,22	9,14	Open	Open	---	---	---	
---	Setup	9,11,12	13,14	Open	Open	---	---	---	

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXVII. Disable/enable recovery time test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
BB17	IE to M0 (t_{PHZ}) Disable/enable recovery time, C_L = 50 pF	9,11,12	13	14	16	6	70	ns	29,30
BB18	$\overline{IE}$ to M0 (t_{PZH})				16	5	57		
BB19	$\overline{IE}$ to M1 (t_{PHZ})				18	6	70		
BB20	$\overline{IE}$ to M1 (t_{PZH})				18	5	57		
BB21	$\overline{IE}$ to M2 (t_{PHZ})				20	6	70		
BB22	$\overline{IE}$ to M2 (t_{PZH})				20	5	57		
BB23	$\overline{IE}$ to M3 (t_{PHZ})				39	6	70		
BB24	$\overline{IE}$ to M3 (t_{PZH})				39	5	57		
BB25	$\overline{IE}$ to M4 (t_{PHZ})				6	6	70		
BB26	$\overline{IE}$ to M4 (t_{PZH})				6	5	57		
BB27	$\overline{IE}$ to M5 (t_{PHZ})				4	6	70		
BB28	$\overline{IE}$ to M5 (t_{PZH})				4	5	57		
BB29	$\overline{IE}$ to M6 (t_{PHZ})				2	6	70		
BB30	$\overline{IE}$ to M6 (t_{PZH})				2	5	57		
BB31	$\overline{IE}$ to M7 (t_{PHZ})				22	6	70		
BB32	IE to M7 (t_{PZH})				22	5	57		
---	Setup	Open	9,11,12,13,14	Open	Open	---	---	---	
---	Setup	9,13	11,12,14,36,37, 28,29,31	Open	Open	---	---	---	
---	Setup	11,12	9,13,14,36,37	Open	Open	---	---	---	

Source: https://assist.dla.mil -- Downloaded: 2026-02-20 17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXVII. Disable/enable recovery time test - Continued.

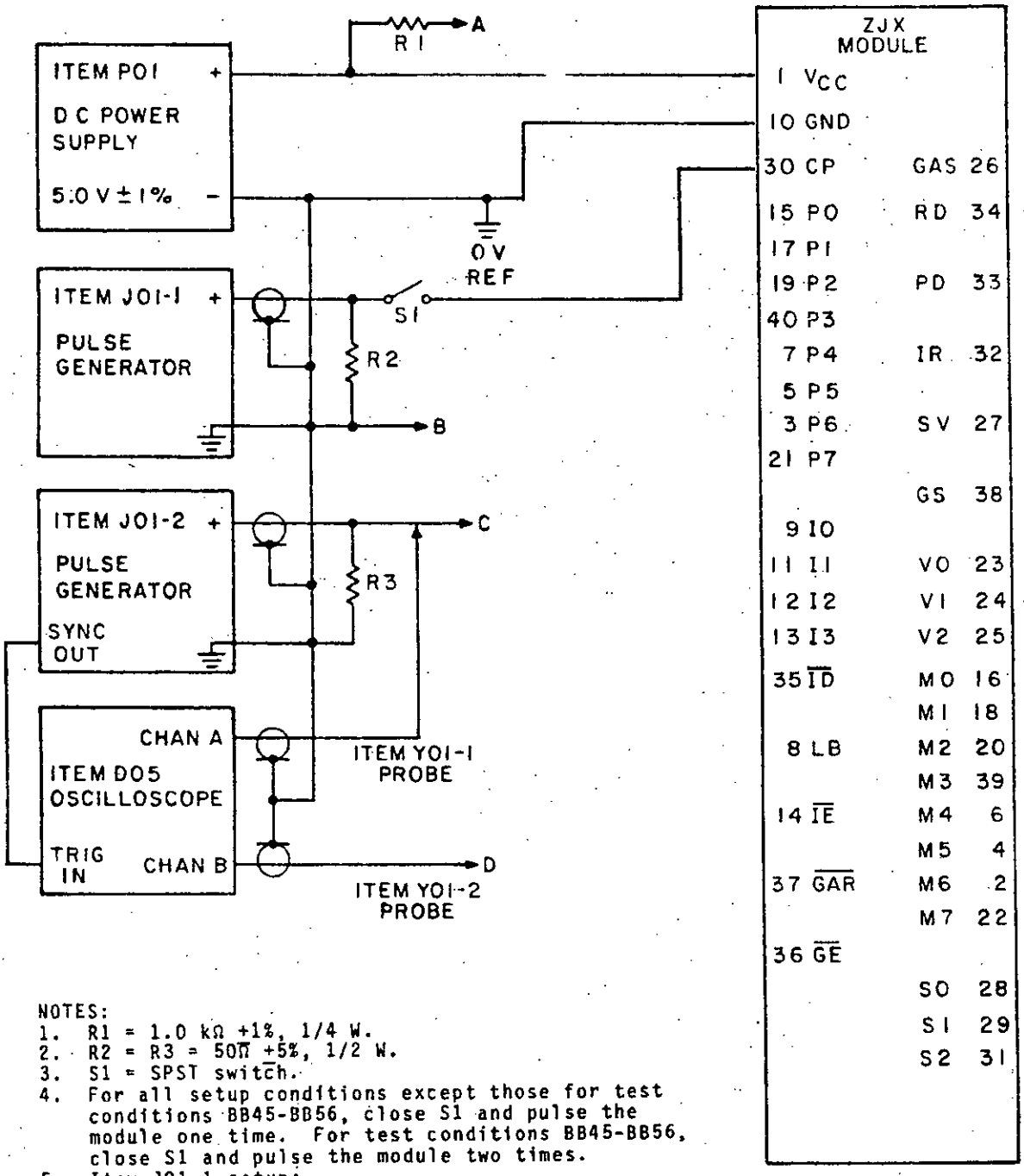
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
BB33	IE to S0 (t_{PLZ}) Disable/enable recovery time, C_L = 50 pF	11,12	9,13,36,37	14	28	5	59	ns	29,30
BB34	IE to S0 (t_{PZL})				28				
BB35	IE to S1 (t_{PLZ})				29				
BB36	IE to S1 (t_{PZL})				29				
BB37	IE to S2 (t_{PLZ})				31				
BB38	IE to S2 (t_{PZL})				31				
---	Setup	Open	9,11,12,13,14	Open	Open	---	---	---	
---	Setup	9,13,28,29,31	11,12,14,36,37	Open	Open	---	---	---	
---	Setup	11,12	9,13,14,36,37	Open	Open	---	---	---	
BB39	IE to S0 (t_{PHZ}) Disable/enable recovery time, C_L = 50 pF		9,13,36,37	14	28	6	65	ns	
BB40	IE to S0 (t_{PZH})				28	5	59		
BB41	IE to S1 (t_{PHZ})				29	6	65		
BB42	IE to S1 (t_{PZH})				29	5	59		
BB43	IE to S2 (t_{PHZ})				31	6	65		
BB44	IE to S2 (t_{PZH})				31	5	59		
---	Setup	Open	9,11,12,13,14	Open	Open	---	---	---	
---	Setup	9,12,17,19,40,7,5,3,21	11,13,14,15	Open	Open	---	---	---	
BB45	IE to V0 (t_{PLZ}) Disable/enable recovery time, C_L = 50 pF		11,13,15	14	23	5	59	ns	
BB46	IE to V0 (t_{PZL})				23				
BB47	IE to V1 (t_{PLZ})				24				
BB48	IE to V1 (t_{PZL})				24				
BB49	IE to V2 (t_{PLZ})				25				
BB50	IE to V2 (t_{PZL})				25				

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

* TABLE XXVII. Disable/enable recovery time test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	9,11,12,13,14	Open	Open	---	---	---	29,30
---	Setup	9,12,15,17,19, 40,7,5,3	11,13,14,21	Open	Open	---	---	---	
BB51	IE to VO (t_{PHZ}) Disable/enable recovery time, C_L = 50 pF		11,13,21	14	23	6	65	ns	
BB52	IE to VO (t_{PZH})				23	5	59		
BB53	IE to V1 (t_{PHZ})				24	6	65		
BB54	IE to V1 (t_{PZH})				24	5	59		
BB55	IE to V2 (t_{PHZ})				25	6	65		
BB56	IE to V2 (t_{PZH})				25	5	59		

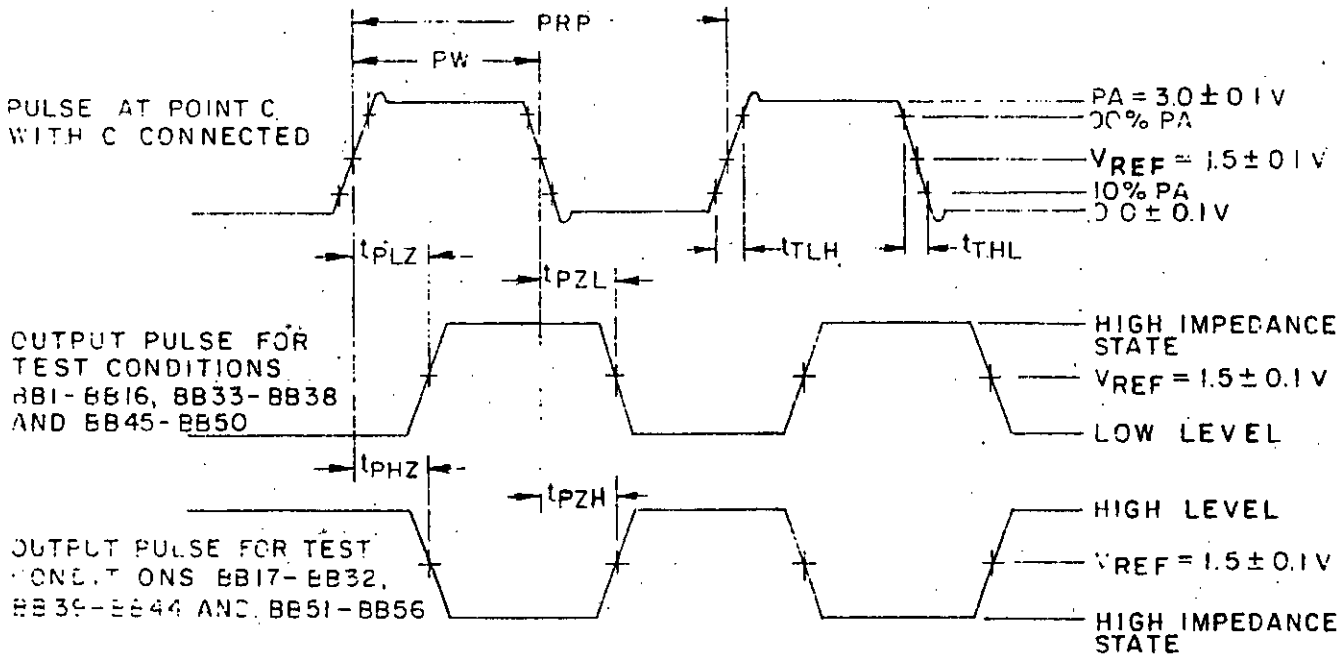
Source: https://assist.dla.mil - Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1.0 kΩ ±1%, 1/4 W.
2. R2 = R3 = 50Ω ±5%, 1/2 W.
3. S1 = SPST switch.
4. For all setup conditions except those for test conditions BB45-BB56, close S1 and pulse the module one time. For test conditions BB45-BB56, close S1 and pulse the module two times.
5. Item J01-1 setup:
   PA = 3.0 ±0.1 V.
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
6. Load all outputs per figure 12.

* FIGURE 29. - Disable/enable recovery time test circuit for table XXVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PW = 500 +50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Before making the measurements, verify that the output low level is $\leq$ 0.5 V and that the high level is $\geq$ 2.4 V.

* FIGURE 30. Disable/enable recovery time test circuit waveforms for table XXVII.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:35Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Naval Electronic Systems Command
ATTN: ELEX 8111
Washington, DC 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/305A(EC)
Controller, Interrupt, Digital, Key Code ZJX

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:35Z
Check the source to verify that this is the current version before use.